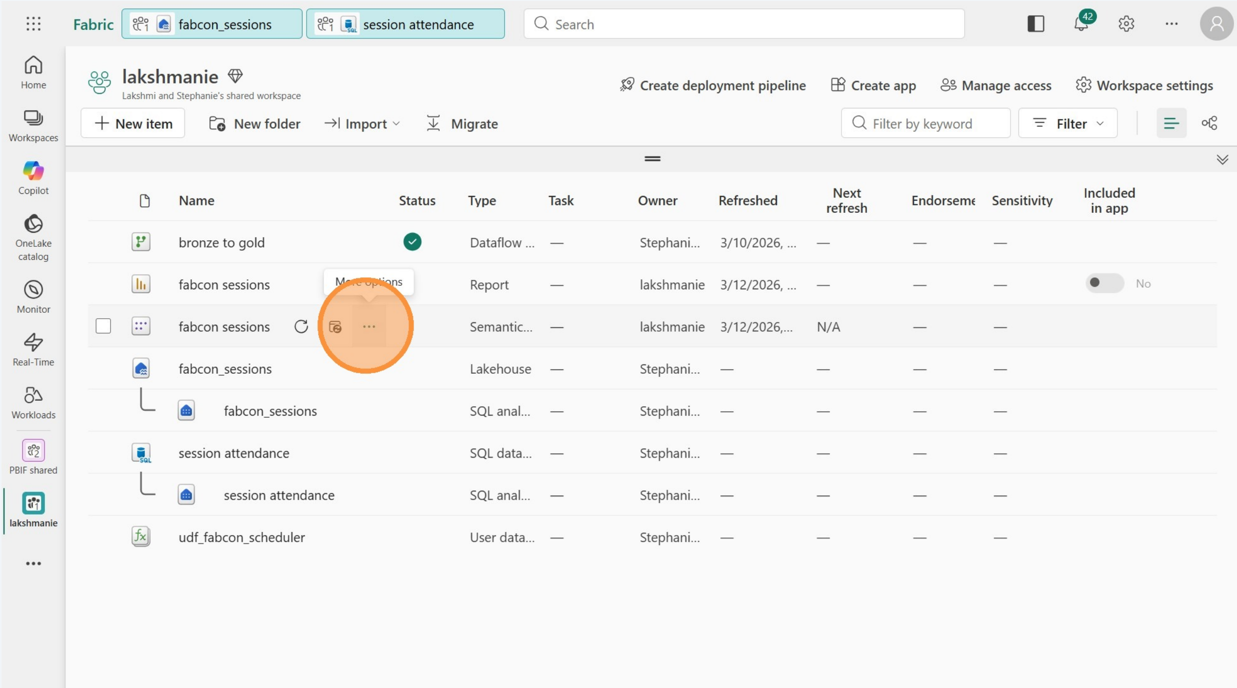
4b. Finish the semantic model in the service

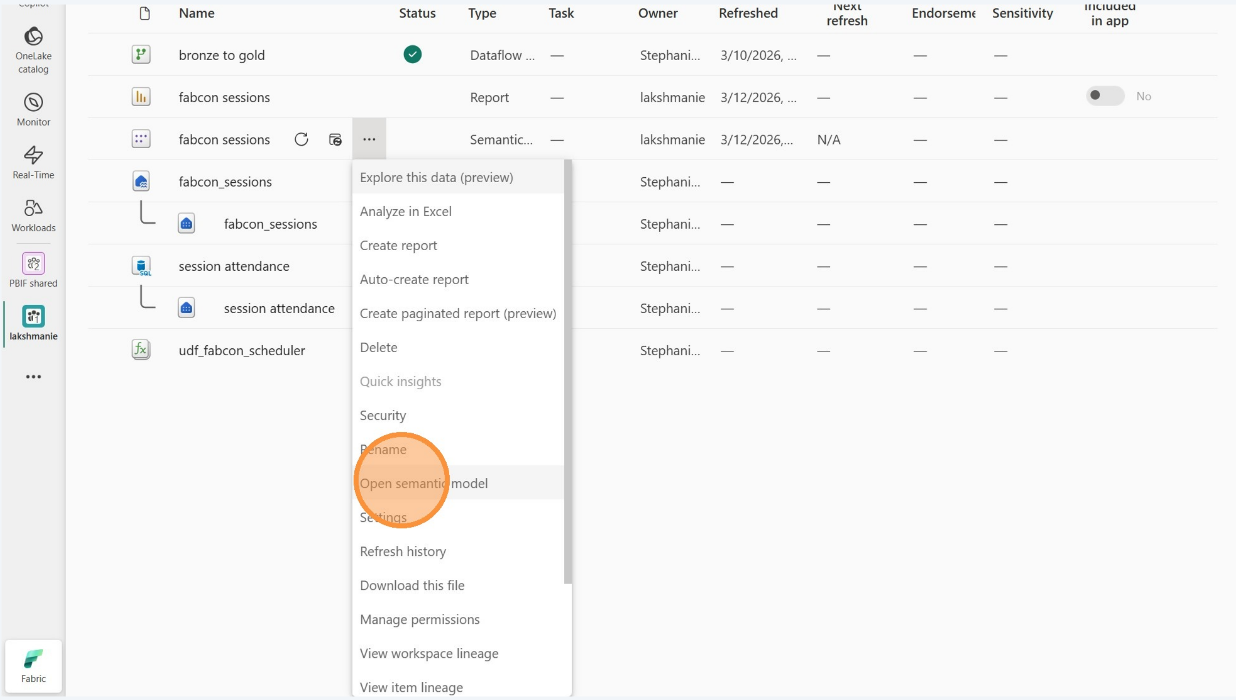
We need to create relationships and measures before we can create the report.

1 Go to your workspace if you're not already there. Hover over the name of the semantic model you just published and click the 3 dots to open the menu.



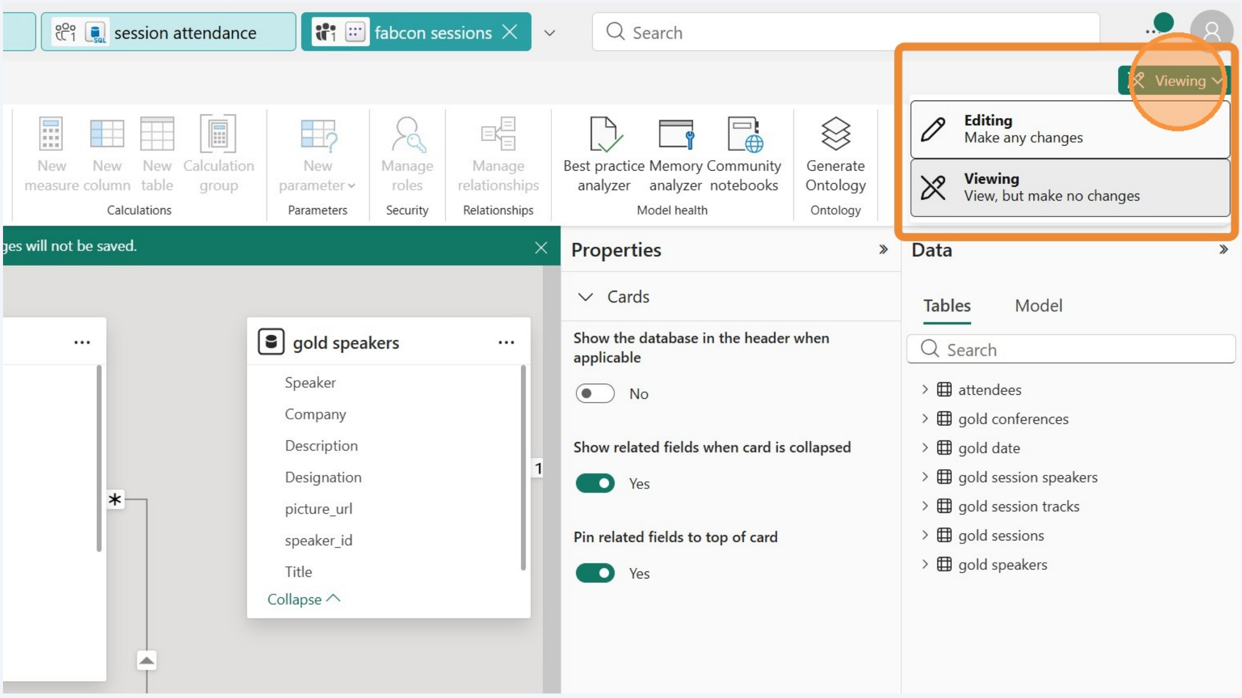
2

Click "Open semantic model." If you accidentally clicked on the name of the semantic model, there is a green "Open" button you can also click.

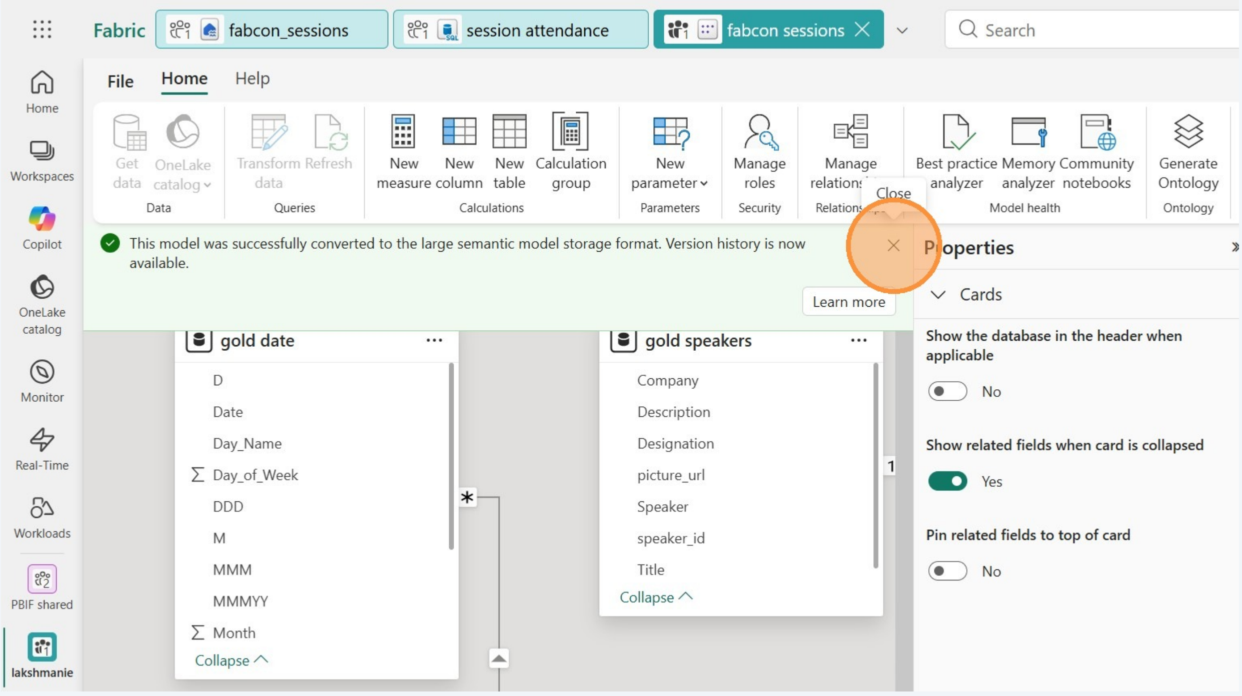


3

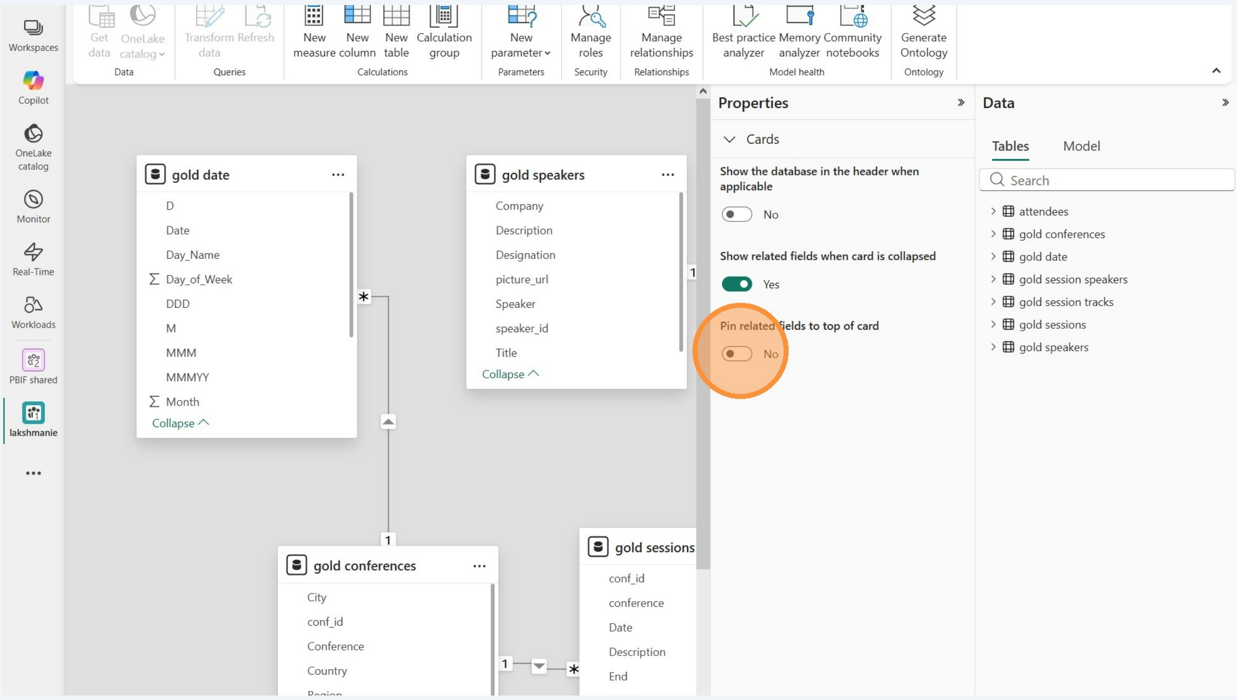
When you first open a semantic model, you're in "Viewing" mode. We need to be in "Editing" mode to make changes. Click the green "Viewing" dropdown in the top-right, and choose "Editing".



4 You will get a message that the model is being updated to the large semantic model storage format. Once it's finished, you can close the below message.

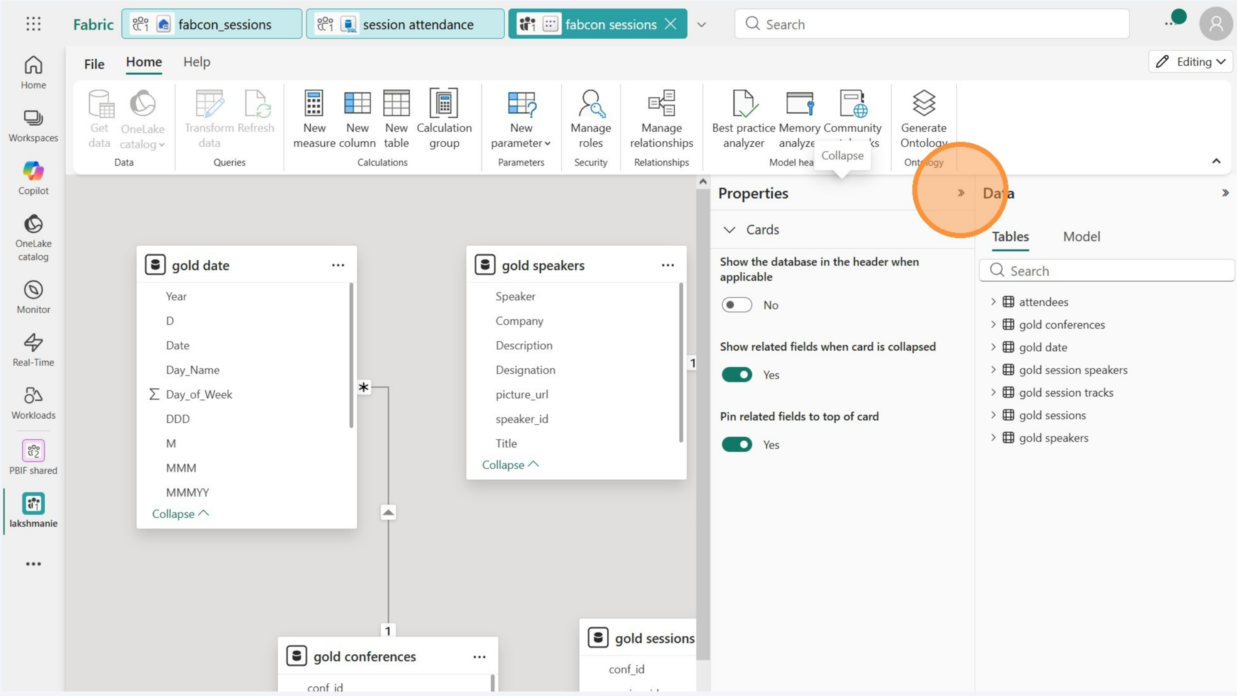


5 Optionally, click the toggle to "Pin related fields to top of card." Not required, but it can help to easily see which columns are in relationships.



6

If your screen is small, you can minimize the Properties pane to see more of the model.

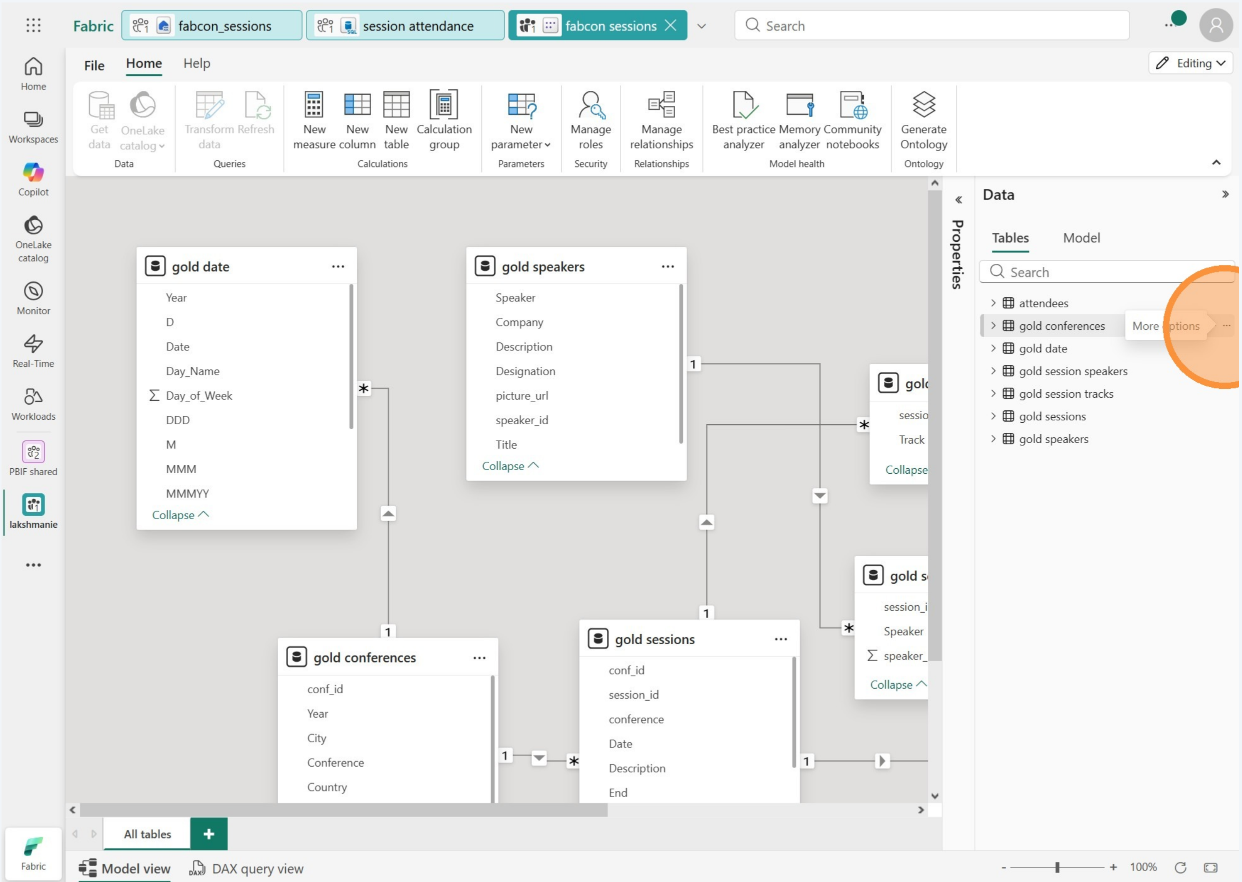


Rename the tables

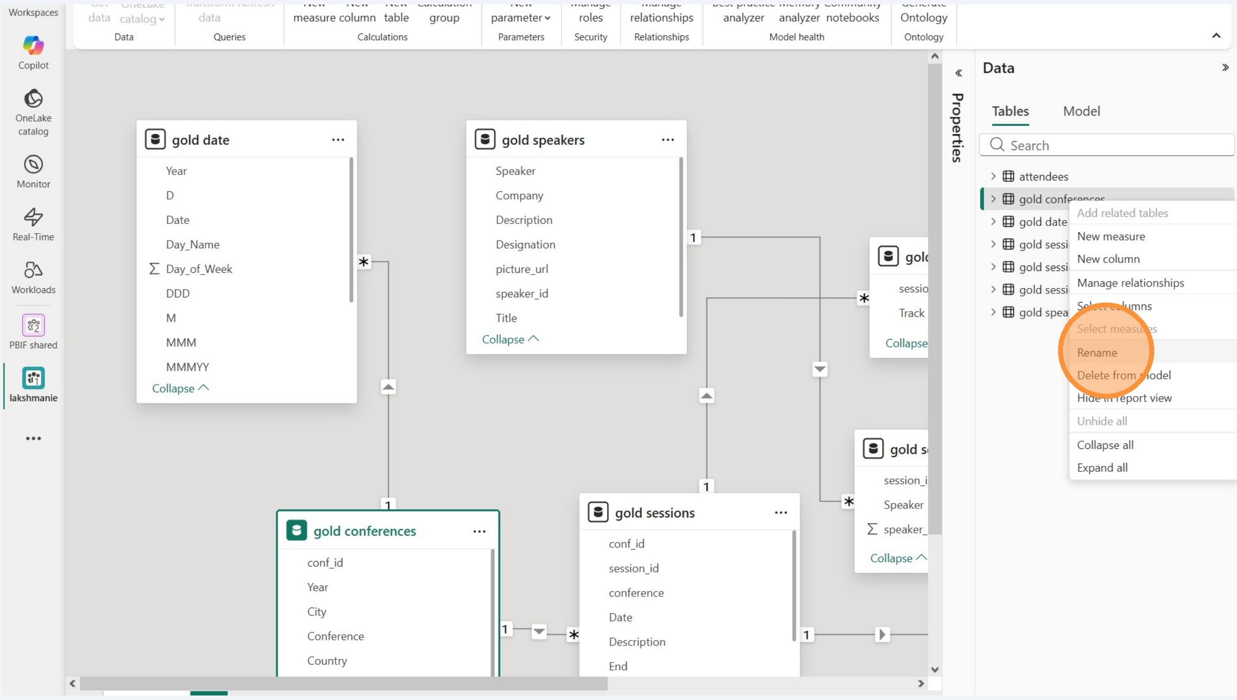
7

Depending on how you imported the tables, you may or may not have the word "gold" as a prefix on some of your tables. It's a good practice to have tables be named so that your colleagues can understand them easily. We're going to rename any of the tables that have "gold " in the beginning.

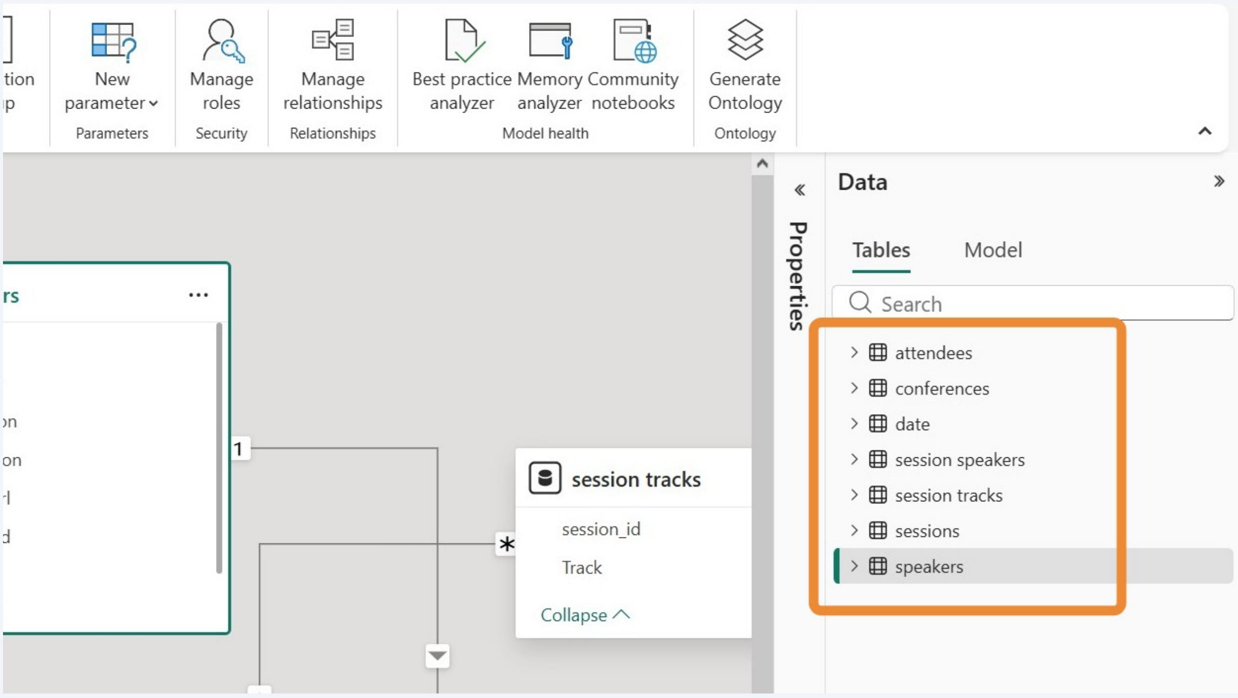
Look at your tables and find one that has "gold " in the beginning of the name. Hover over it and click the 3 dots.



8 Click "Rename" and delete "gold " from the beginning of the name.



9 Repeat the same process to any of your tables that have "gold " as a prefix. Once your done your tables should be named like those below.

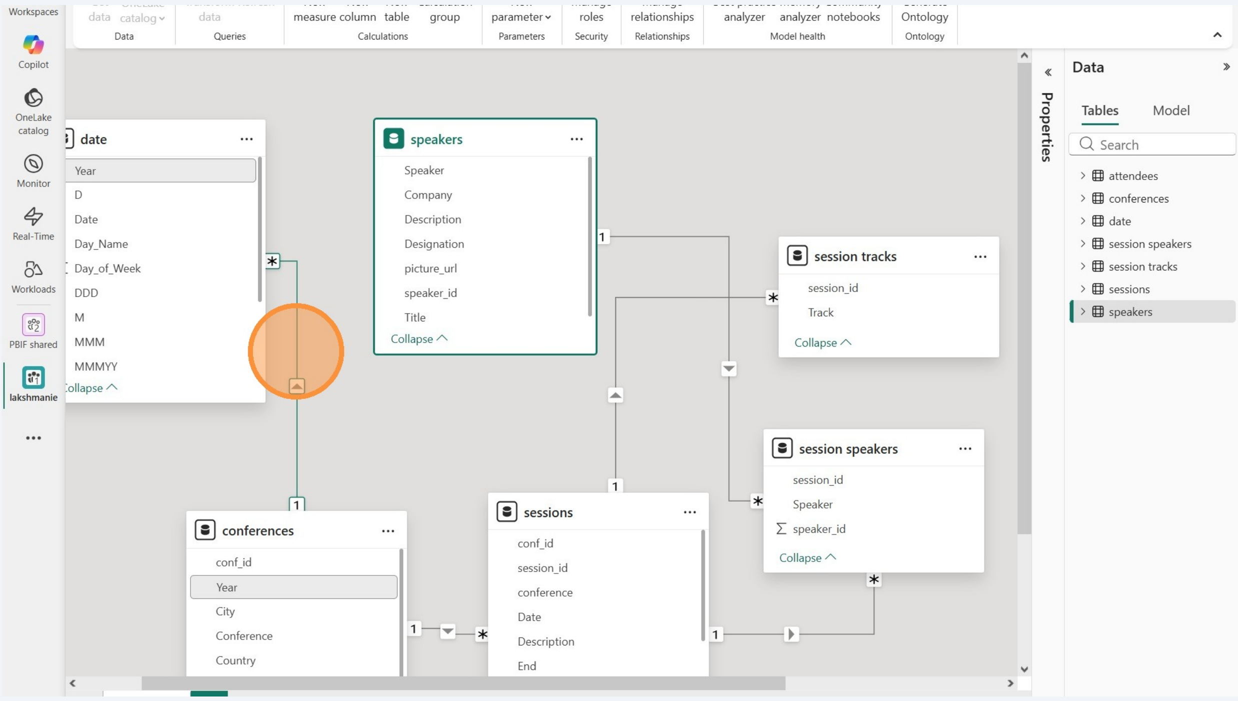


Add relationships

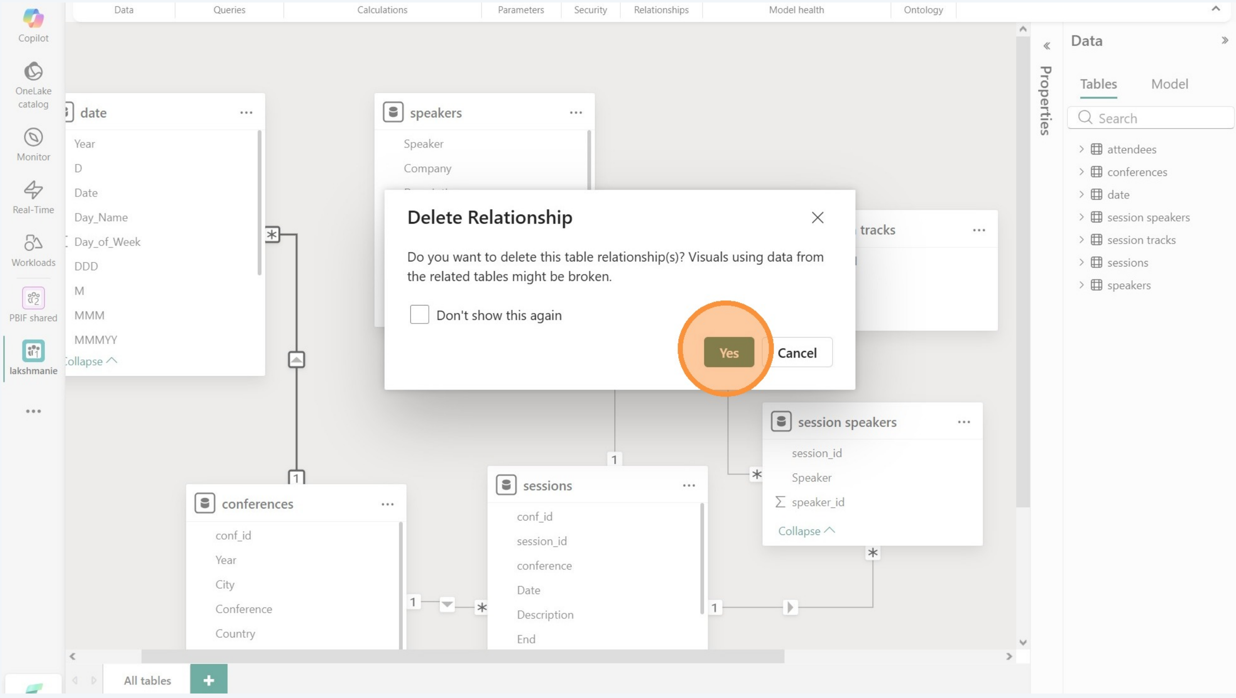
10 We need to edit the auto-created relationships and make sure our tables are properly related. You can rearrange the tables around so you can view them all.

11 First, we don't need a relationship between the date and conferences table so let's delete that.

Right-click on the line connecting "date" to "conferences" and select "Delete."

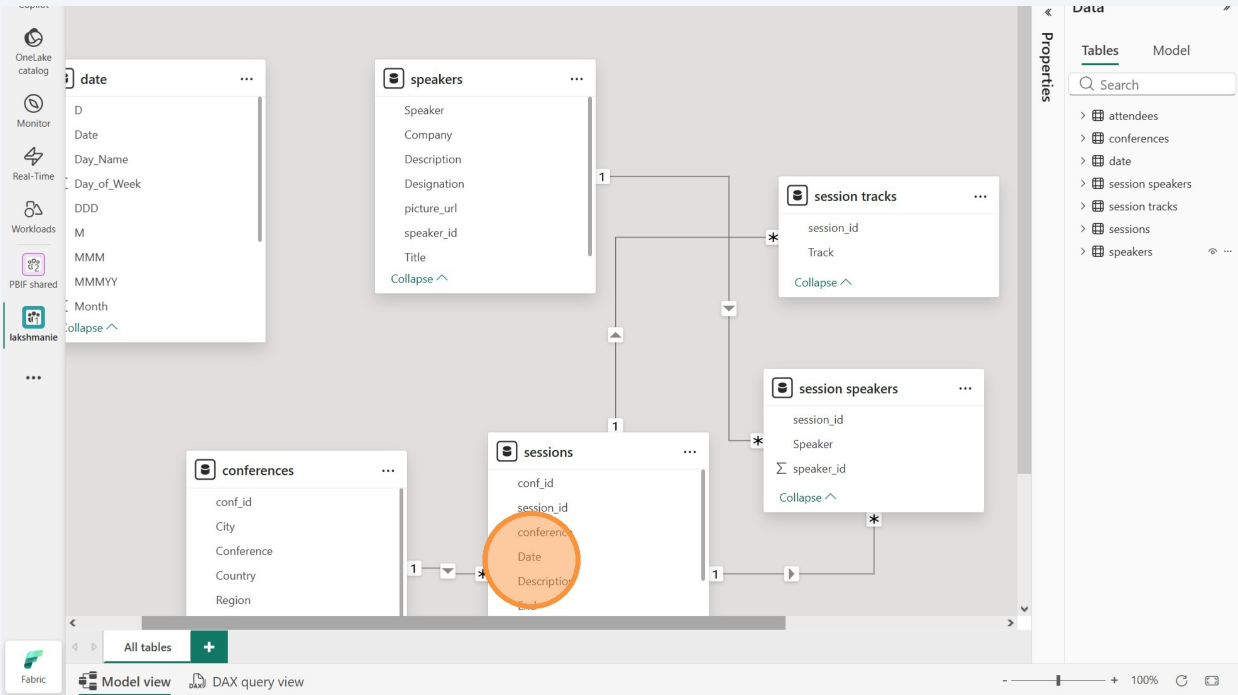


12 Click "Yes"

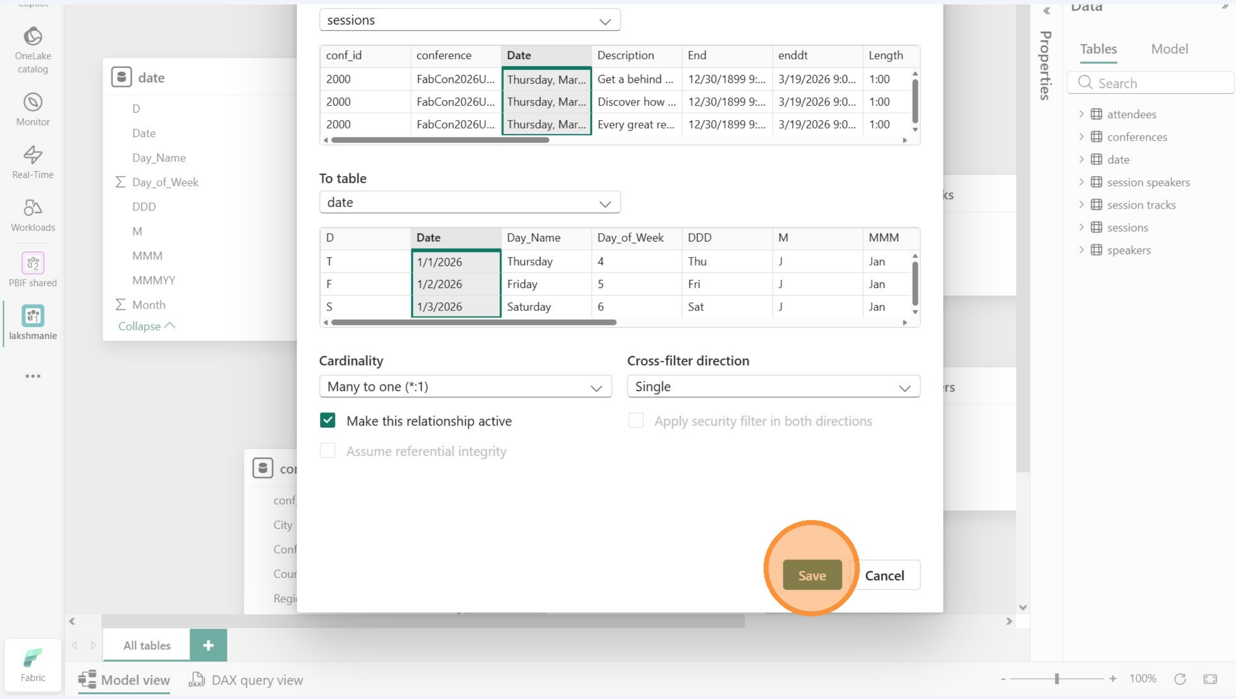


13 We do want to make a relationship between "session" and "date."

Click (and hold down the mouse button) the "Date" column from the "session" table and drag it to the "Date" column of the "date" table. This will open a relationship pop-up window.

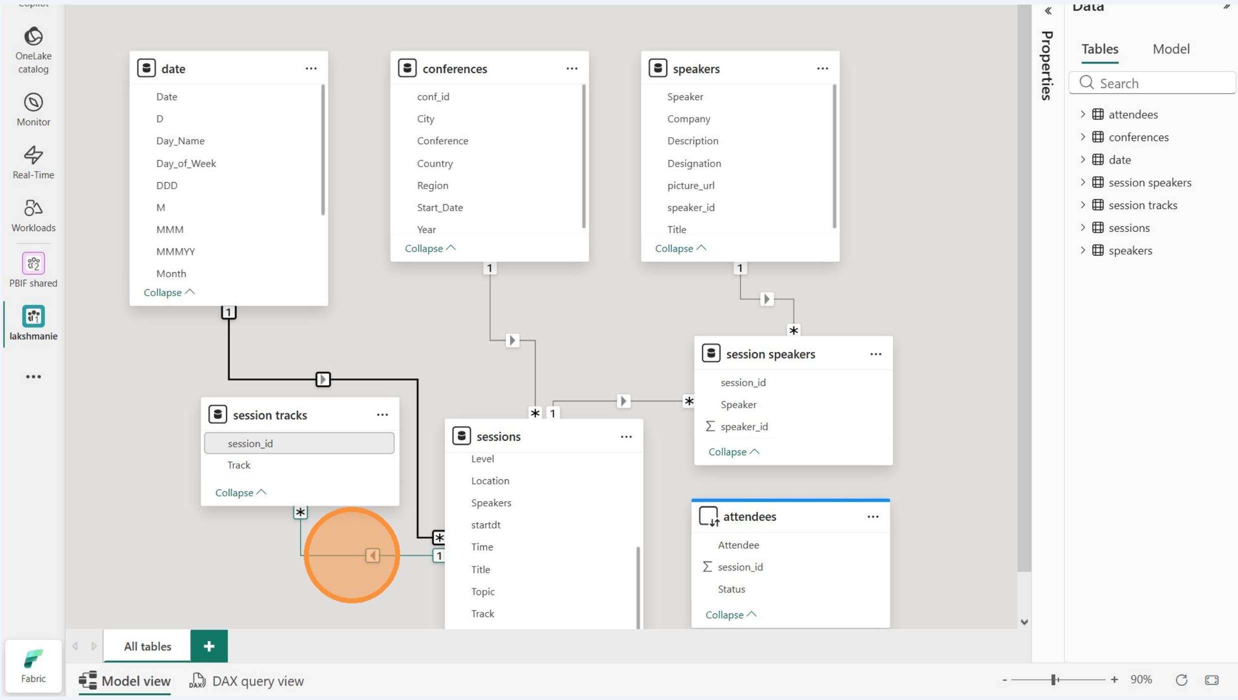


14 The defaults here are correct. There can be one value in the date table with many rows in the session table. This makes sense because each date has multiple sessions. Click "Save."

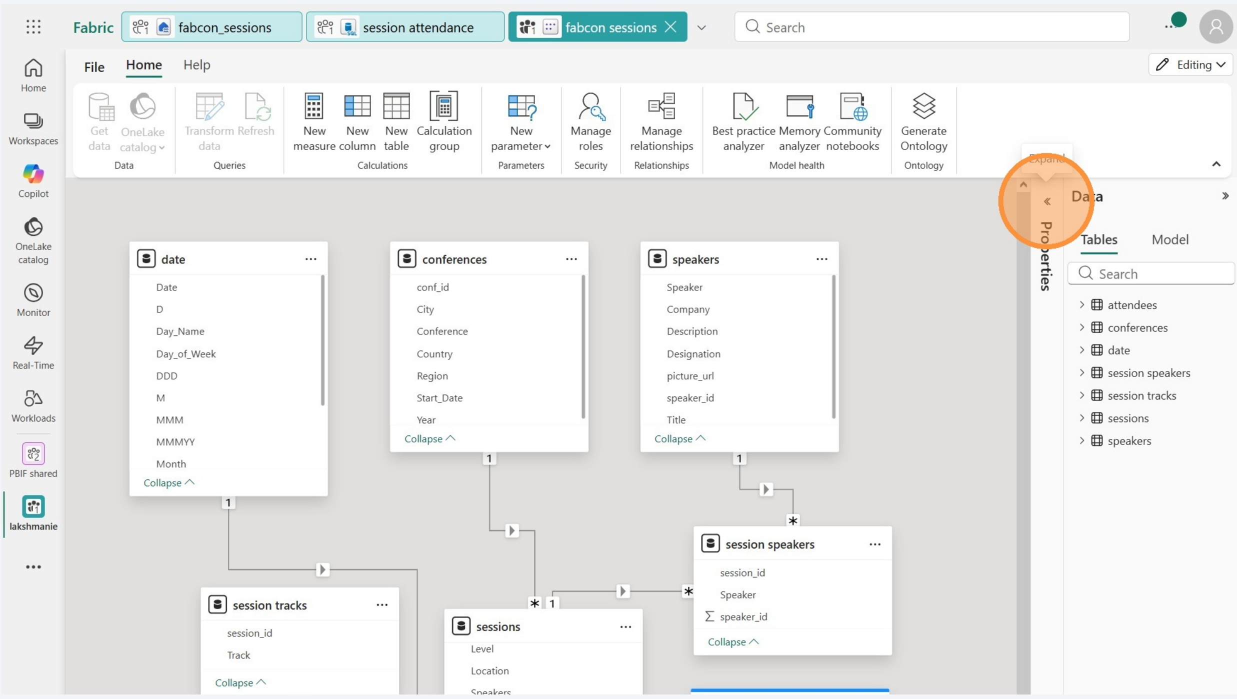


15 The automatically created relationship between sessions and session tracks is almost correct. Each session could have more than one track, and each track. So it does look correct! However, in our reports we would like to have the session tracks filter the sessions - not the other way around. So we want to change the directionality of that relationship.

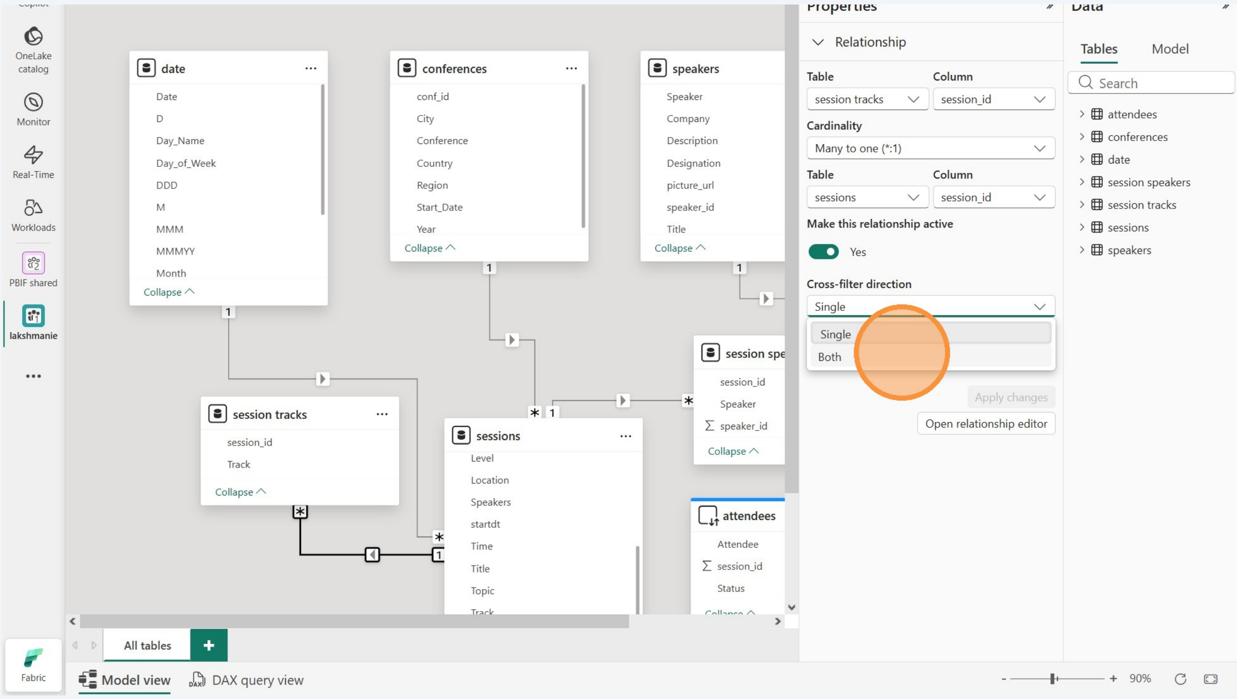
Click on the line connecting sessions to session tracks.



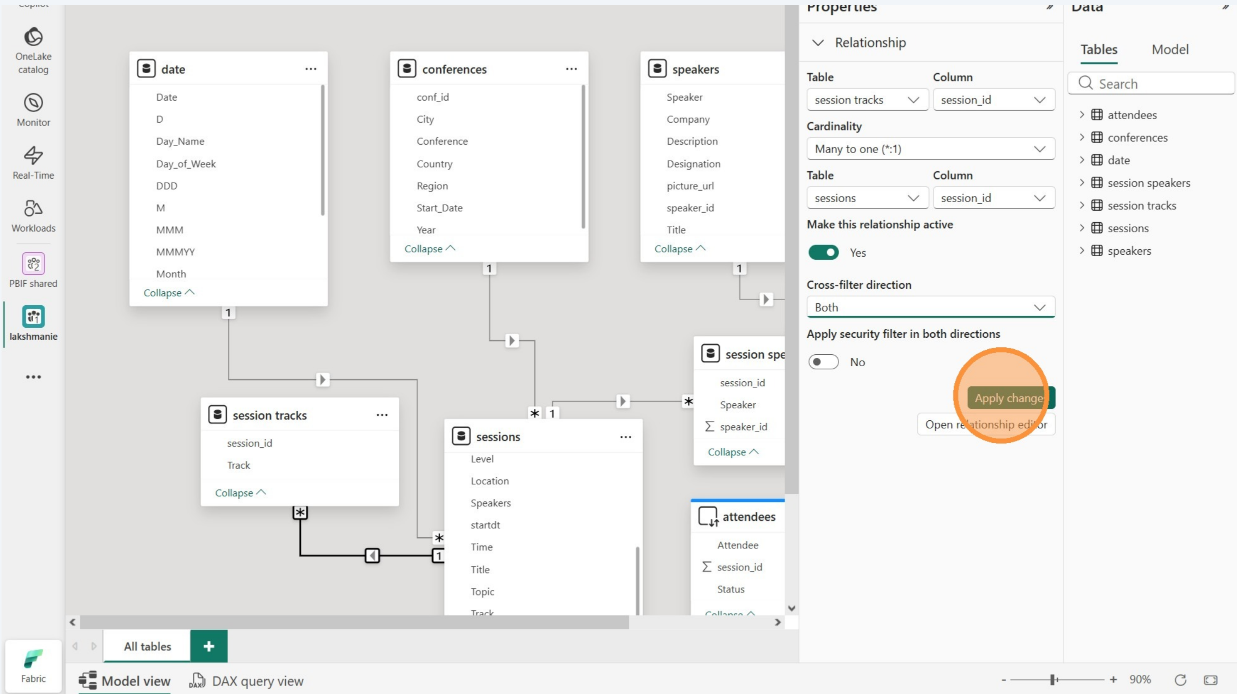
16 If you've got the Properties pane collapsed, expand it.



17 Change the cross-filter direction from Single to Both.



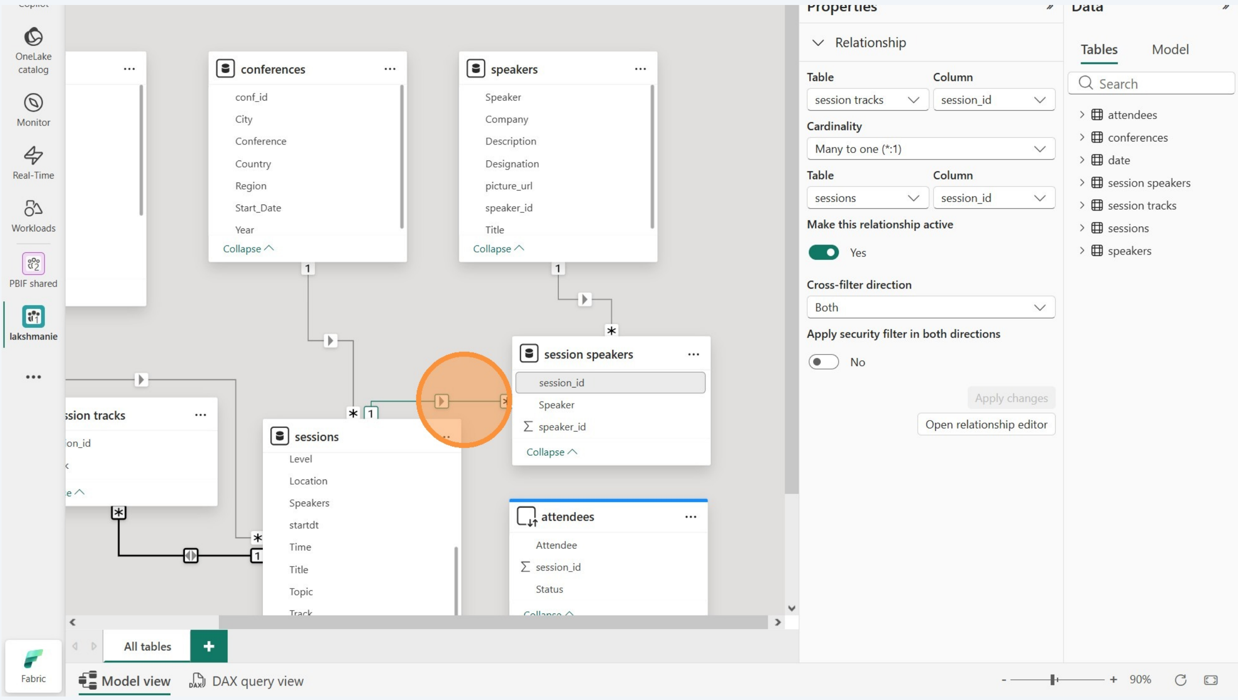
18 Click "Apply changes"



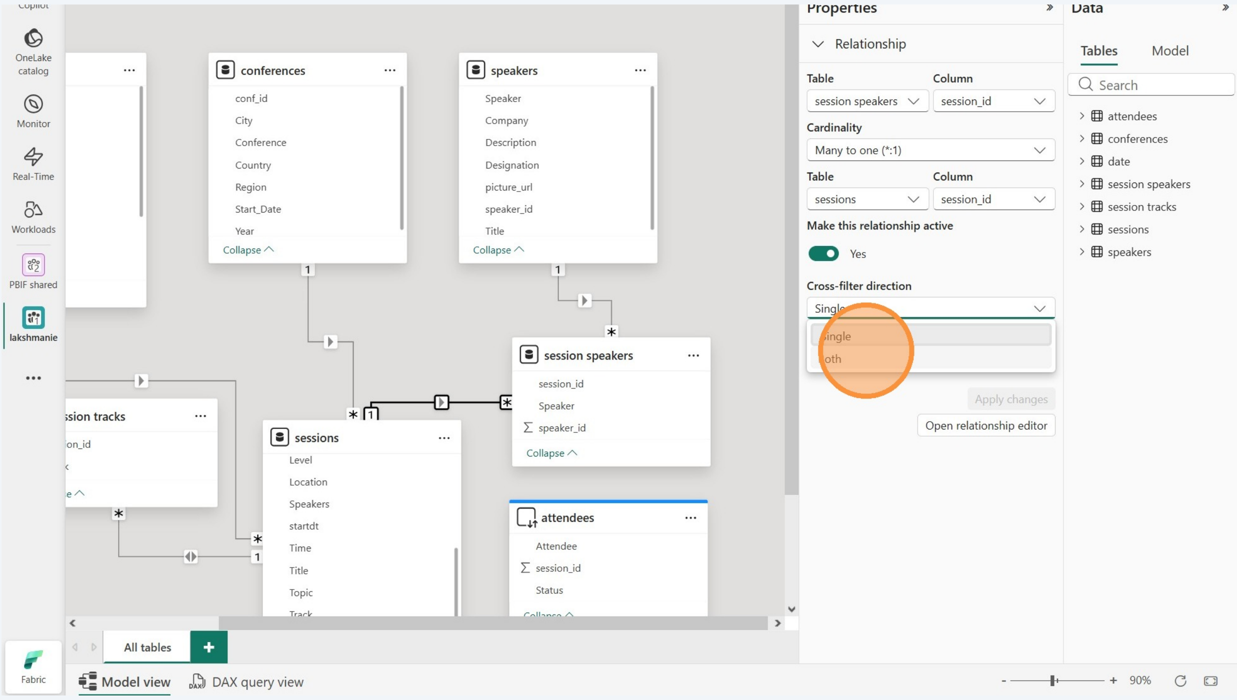
19

The "session speakers" table is what's sometimes referred to as a "bridge" table. We need to be able to have the speakers table filter the sessions those speakers are presenting. But if you look at the directionality of the relationships (which way the arrows are pointing), speakers properly filters "session speakers", but "session speakers" doesn't filter sessions. It's backward. Let's fix that.

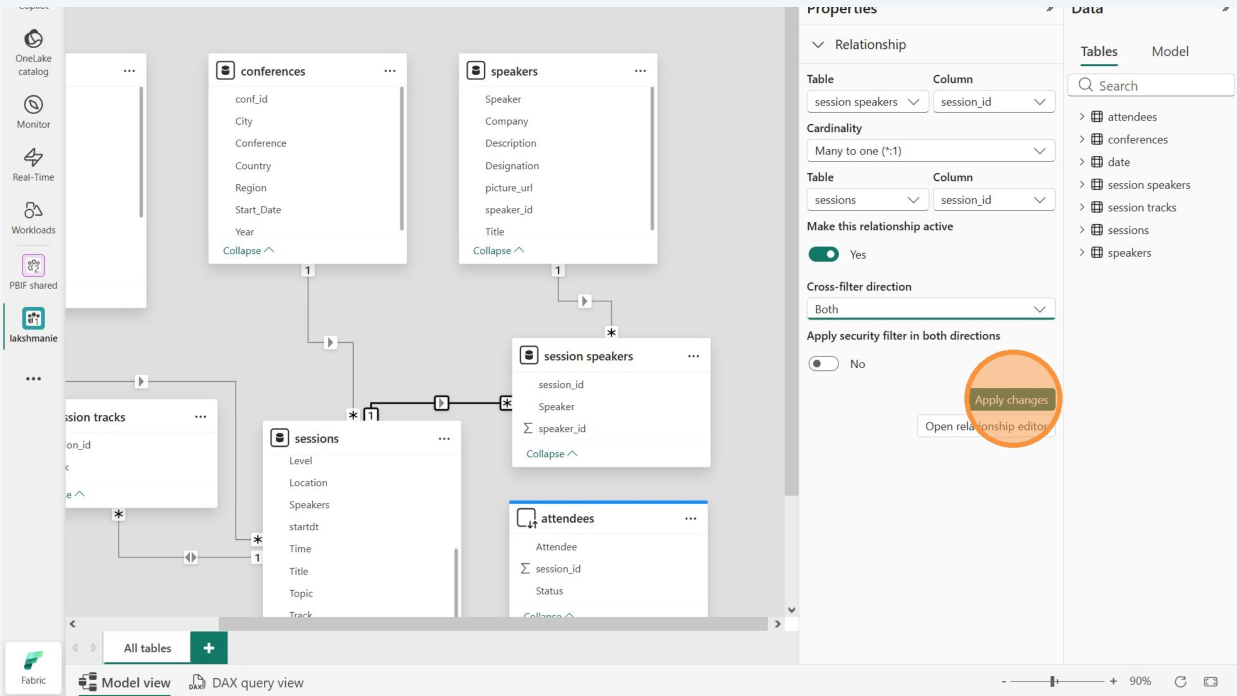
Click on the line connecting sessions to "session speakers."



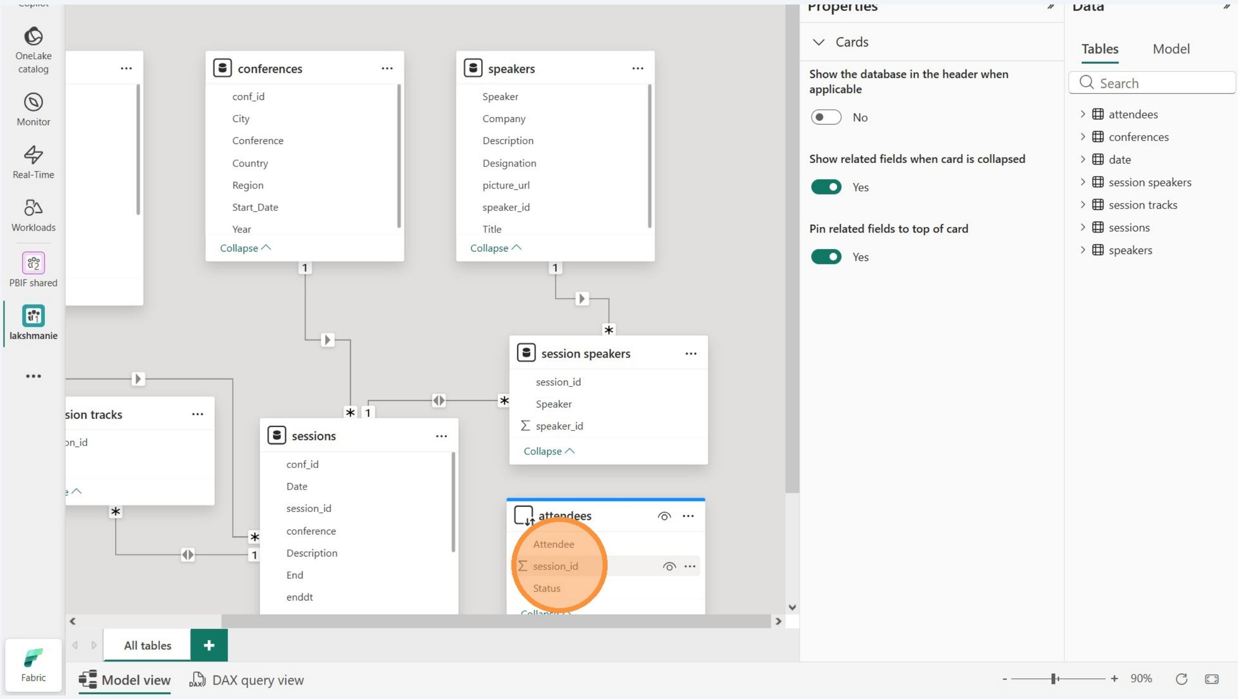
20 Change the relationships Cross-filter direction from "Single" to "Both."



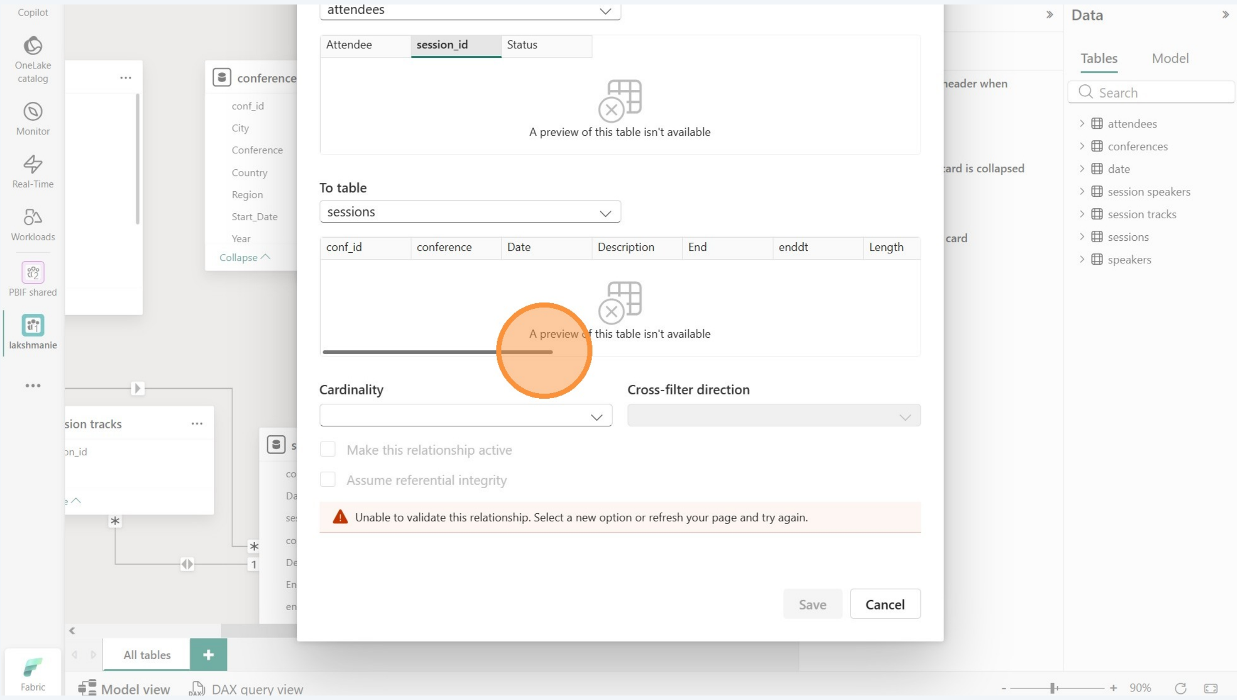
21 Click "Apply changes"



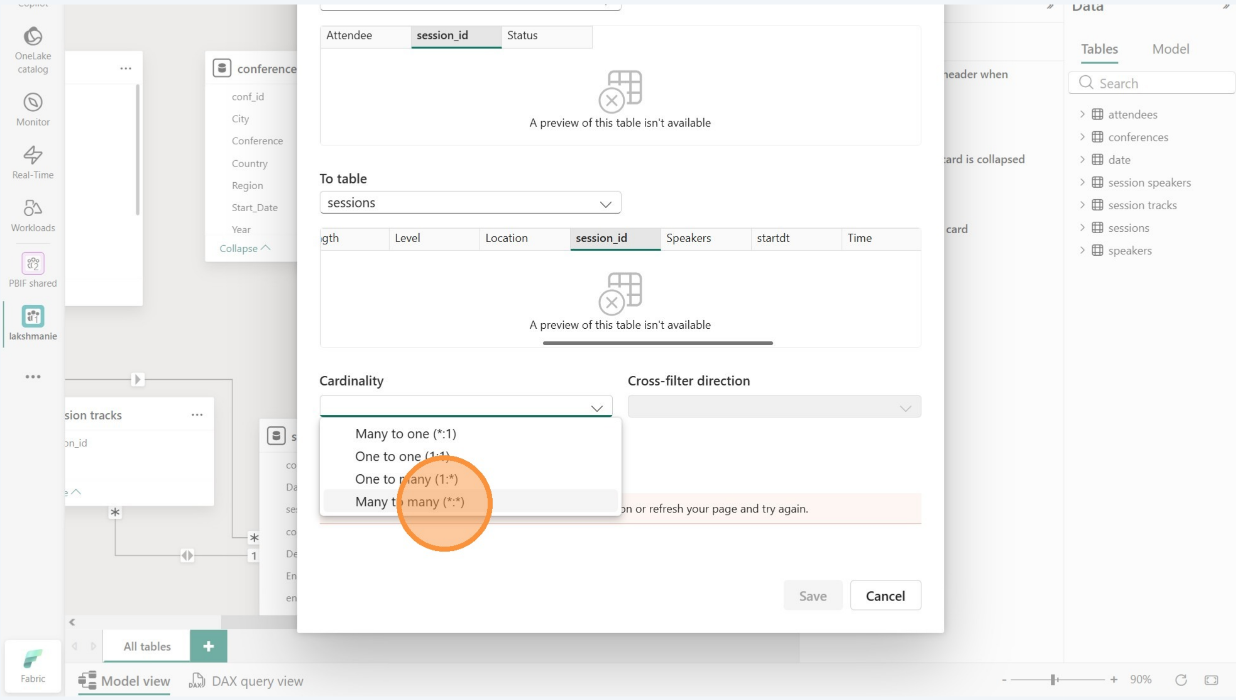
22 The attendees table needs to be related to the sessions table. Grab the "session_id" column from the "attendees" table and drag it over to the "seession_id" column of the "sessions" table.



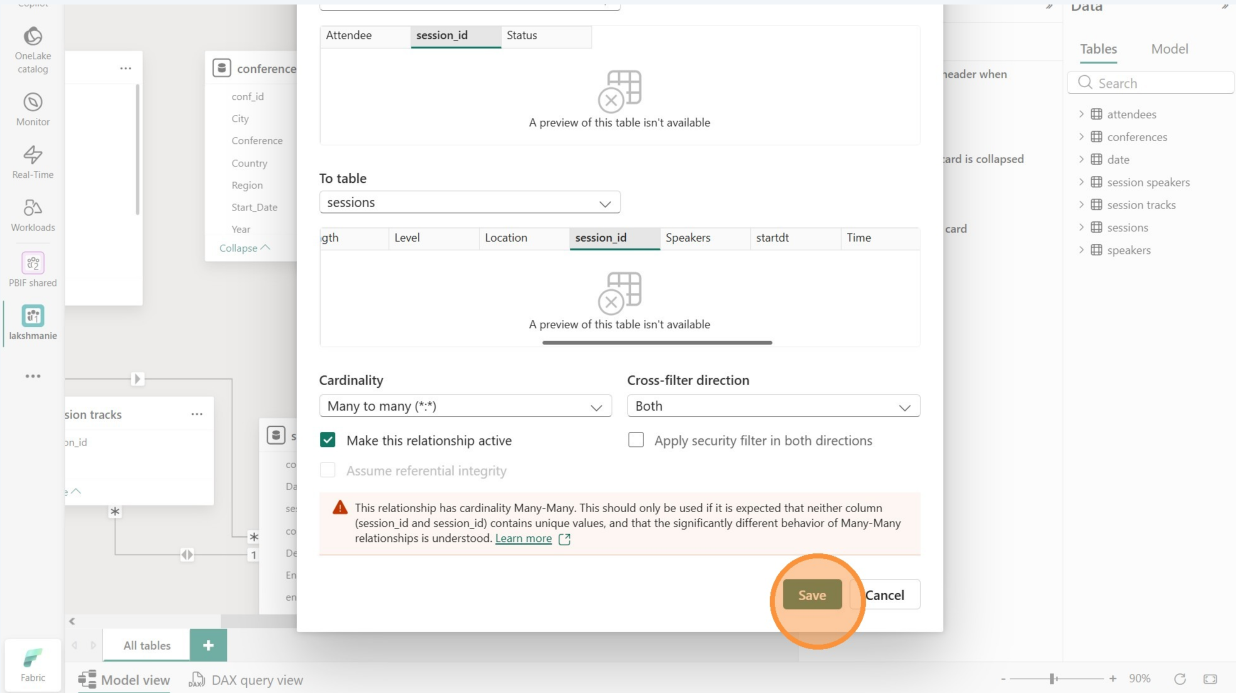
23 The "session_id" column of the attendees table is already selected. We just need to select the "session_id" column in the sessions table. Scroll to the right to find the session_id column. Click the session_id column to select it.



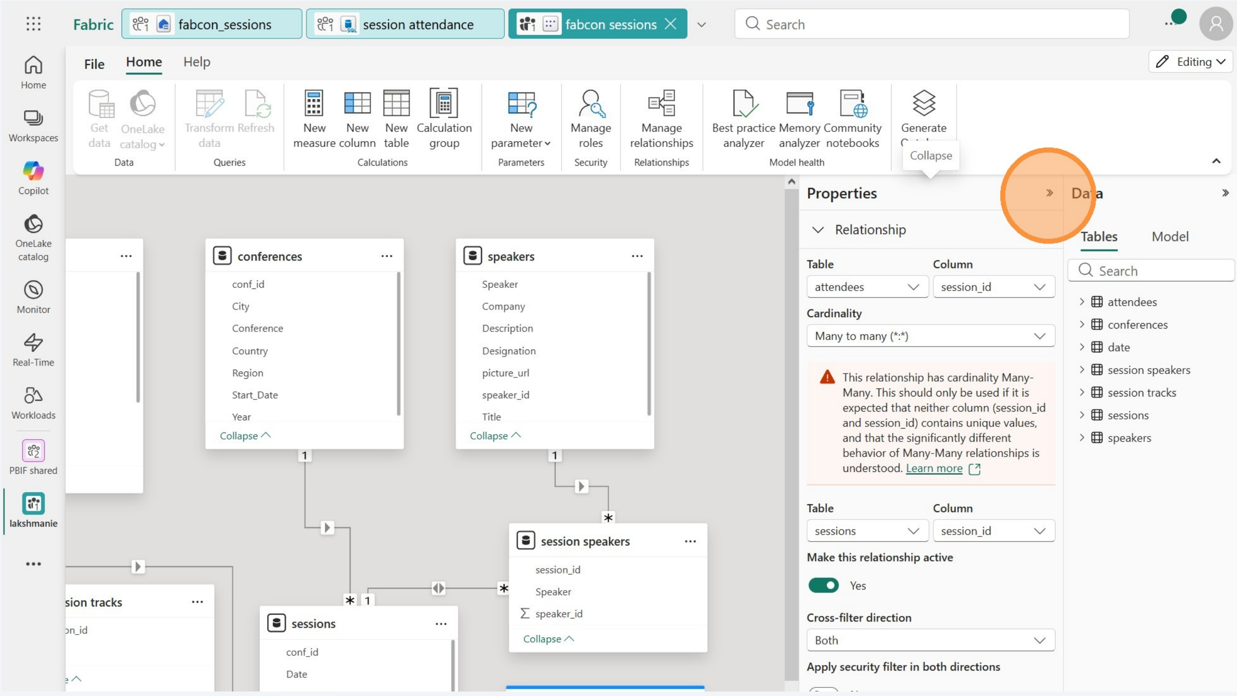
24 Click "Many to many" in the Cardinality dropdown. Each session can have multiple attendees and each attendee can attend multiple sessions.



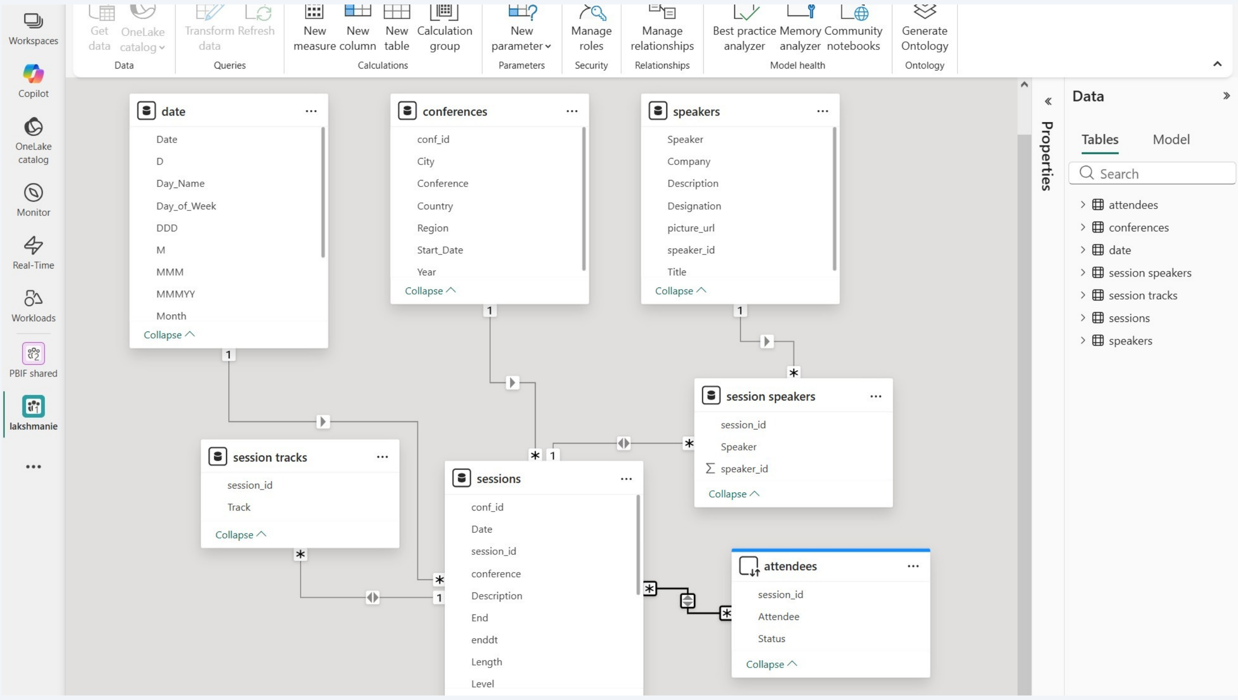
25 Many-to-many relationships can be tricky, so you'll always get a friendly warning when you make them. It's cool. Click "Save."



26 You can minimize the Properties pane again now if you like. Let's make sure our relationships look good.



27 Take a minute to make sure that your relationships look like those below. You can rearrange your tables if it makes it easier.

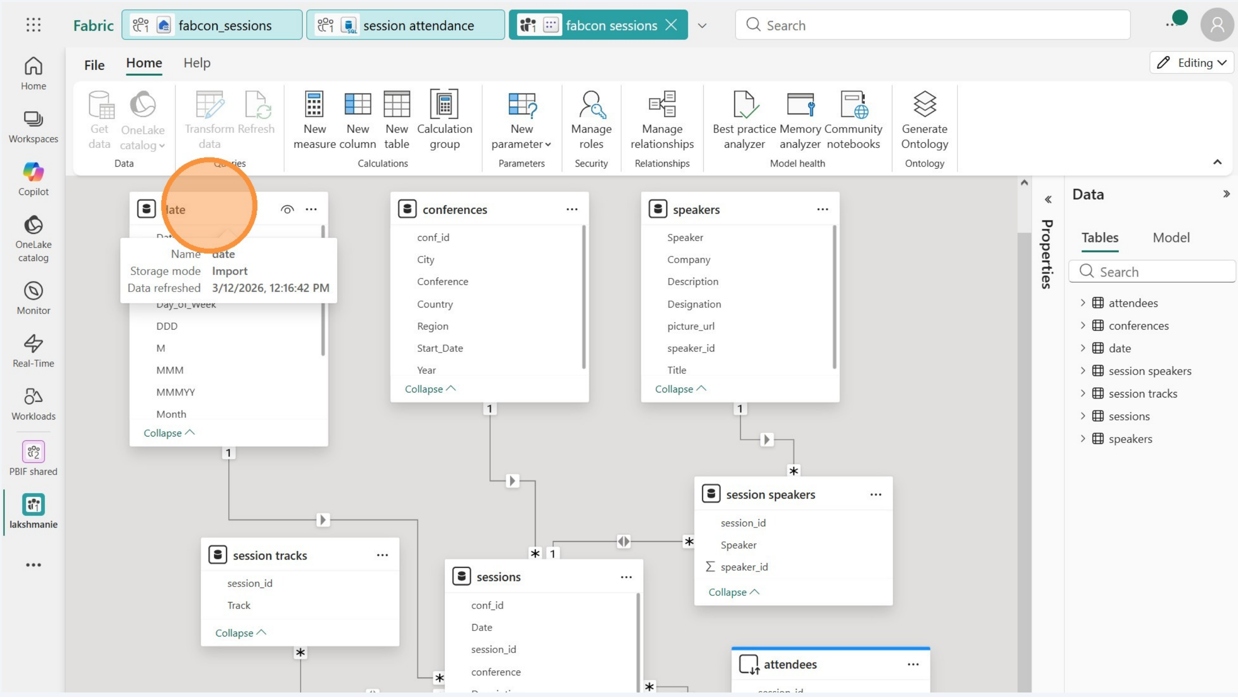


Mark the Date table

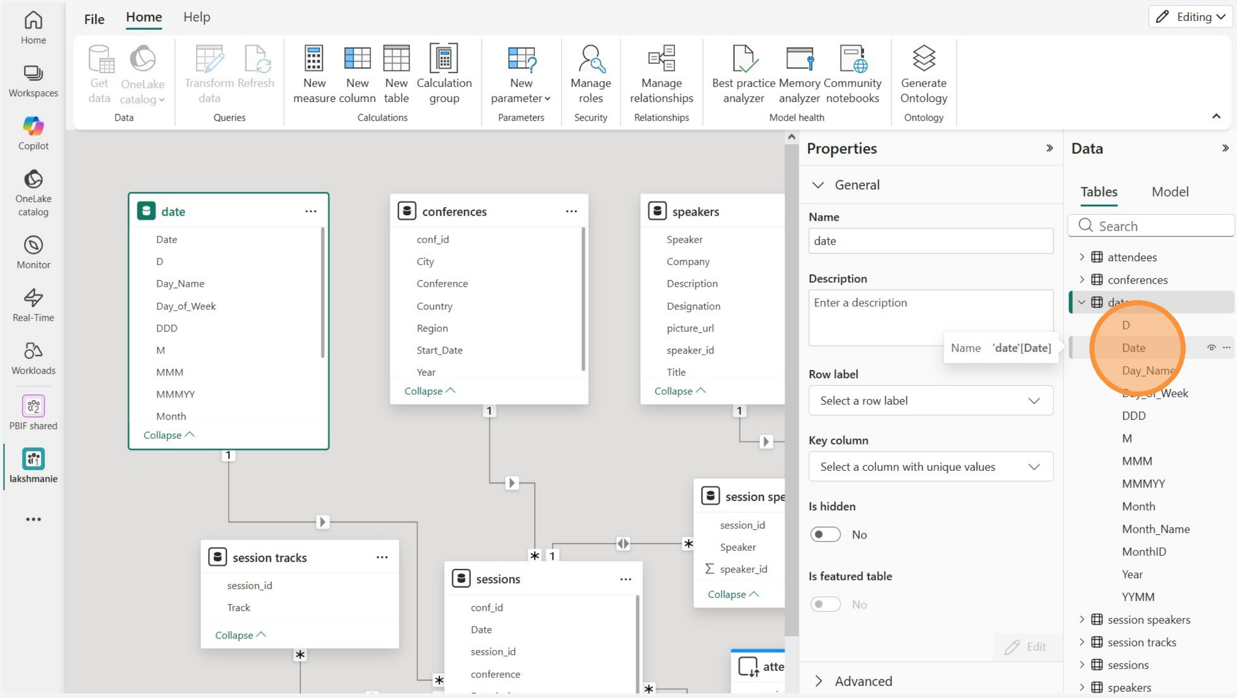
28

To enable time intelligence calculations, it's always a good idea to have a date table and mark it as a date table. BUT - if you don't have a column in a table that is of the date datatype, that won't be possible. So first let's make sure we have a date column.

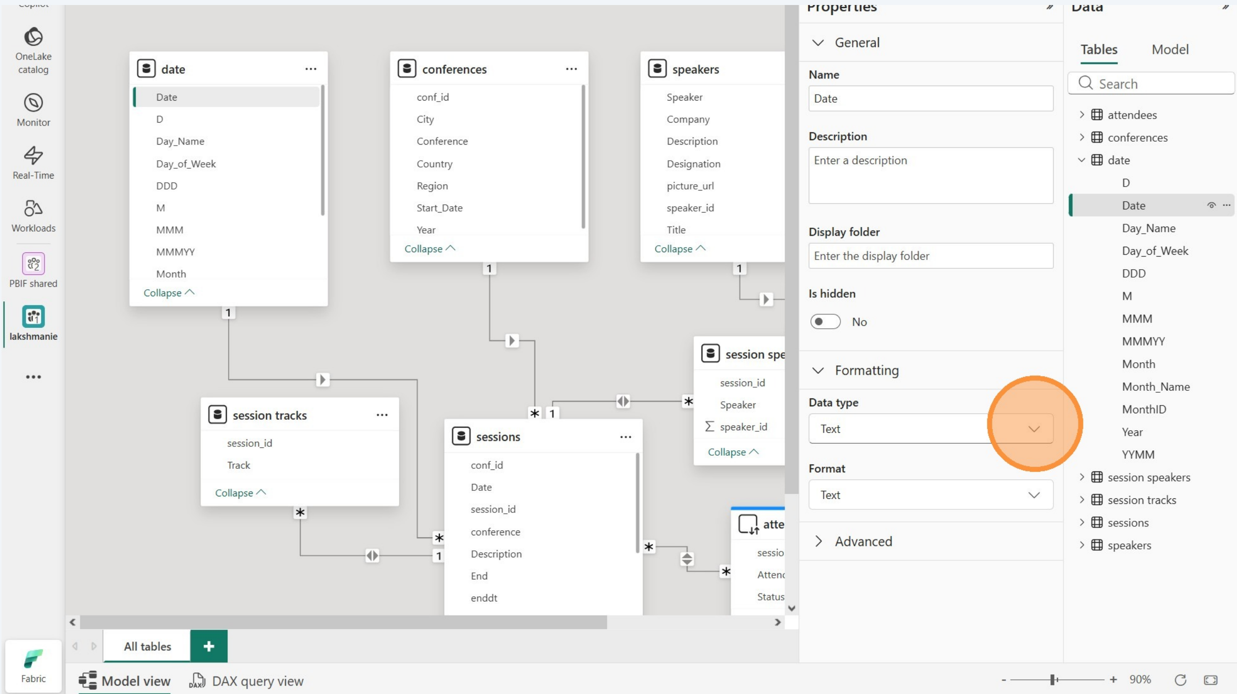
Click the date table to highlight it.



29 Click the "Date" column.

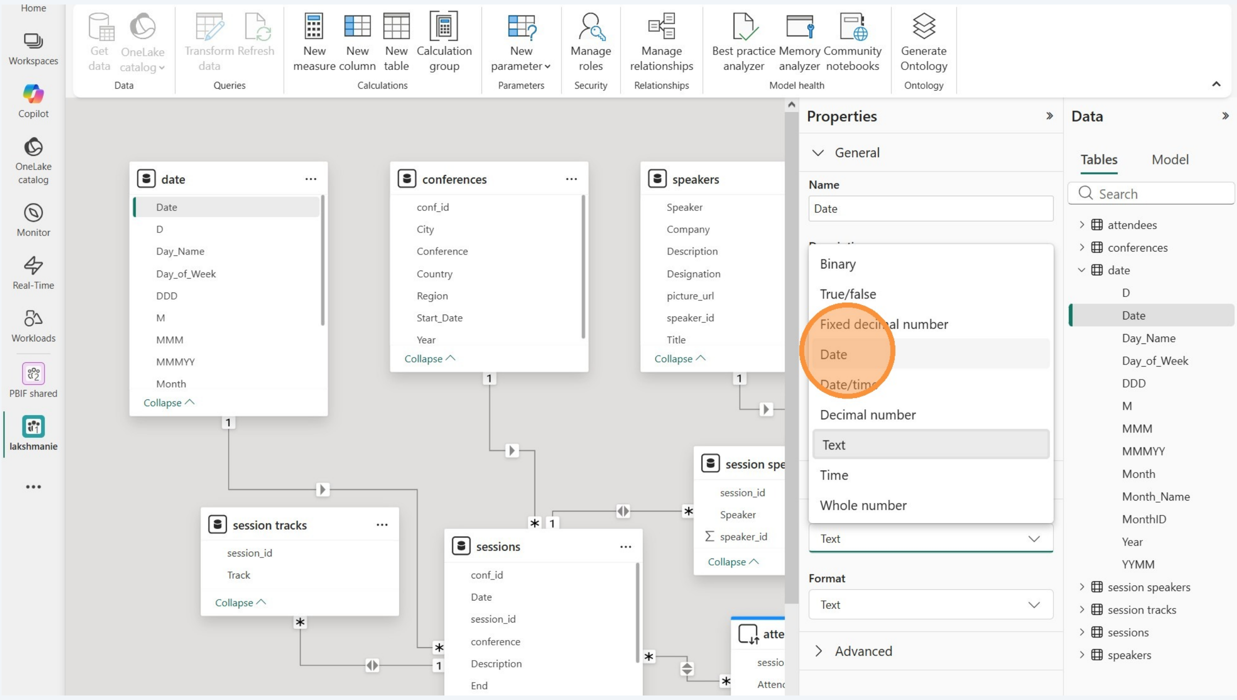


30 The Date column is a "Text" data type. That doesn't sound right. Let's fix that.



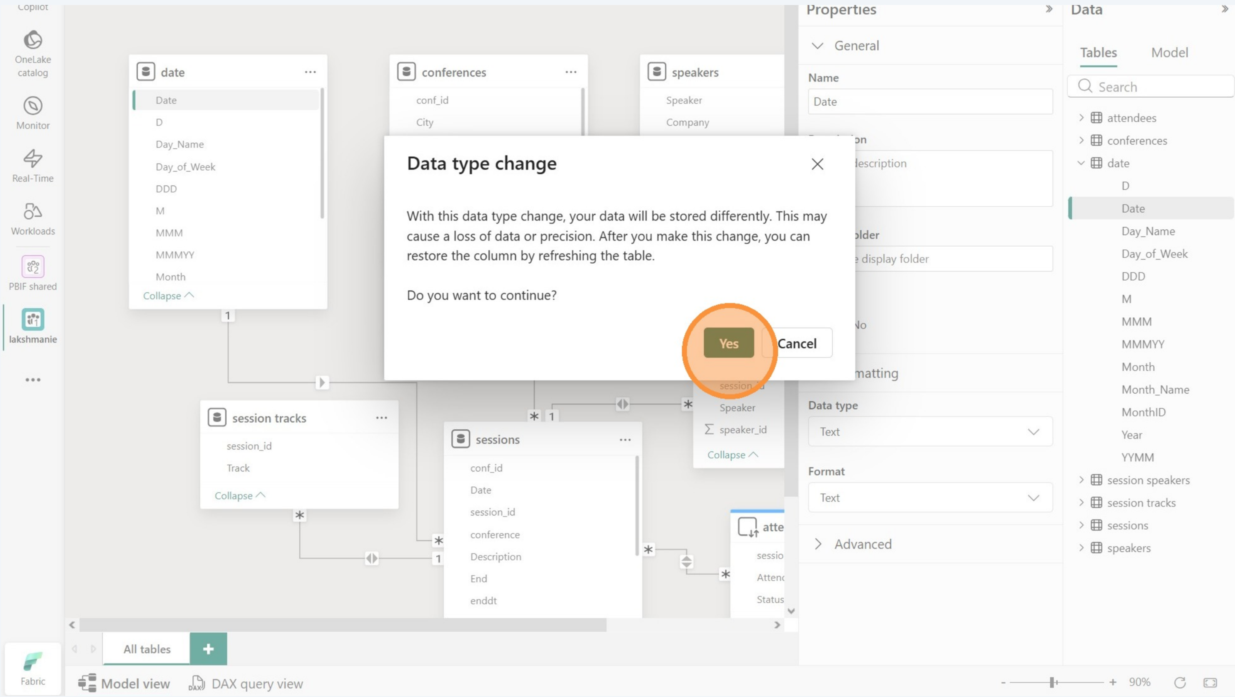
31

Change it to the "Date" data type.

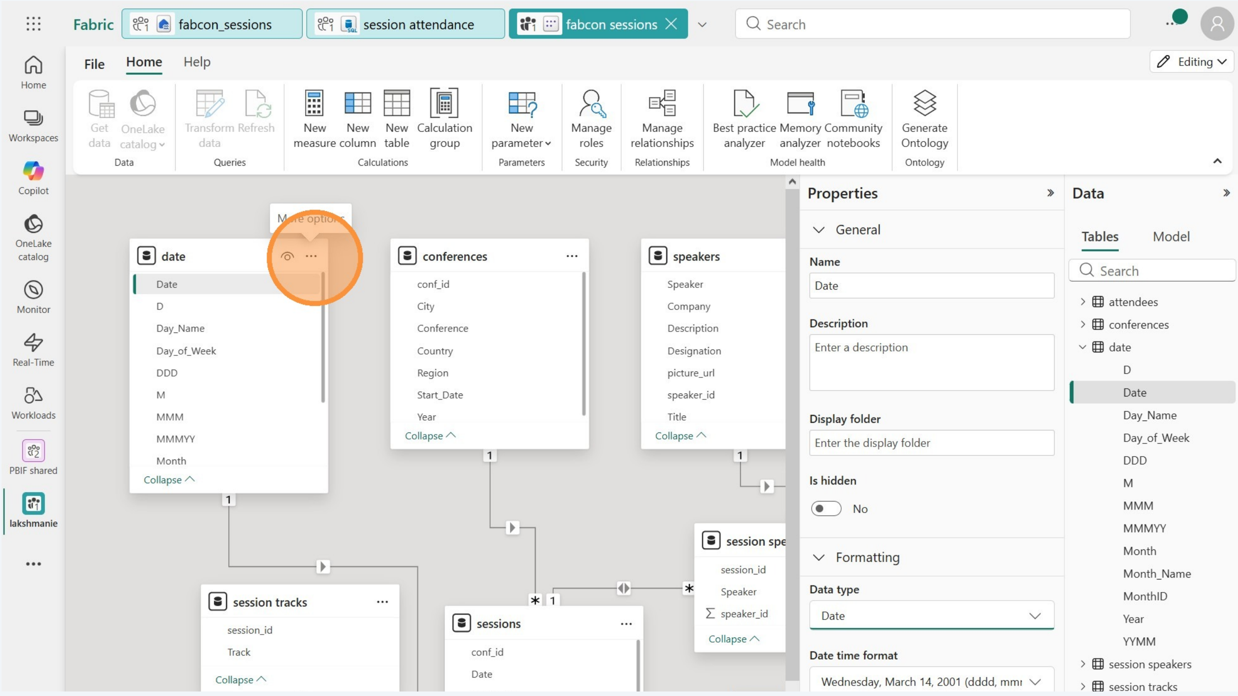


32

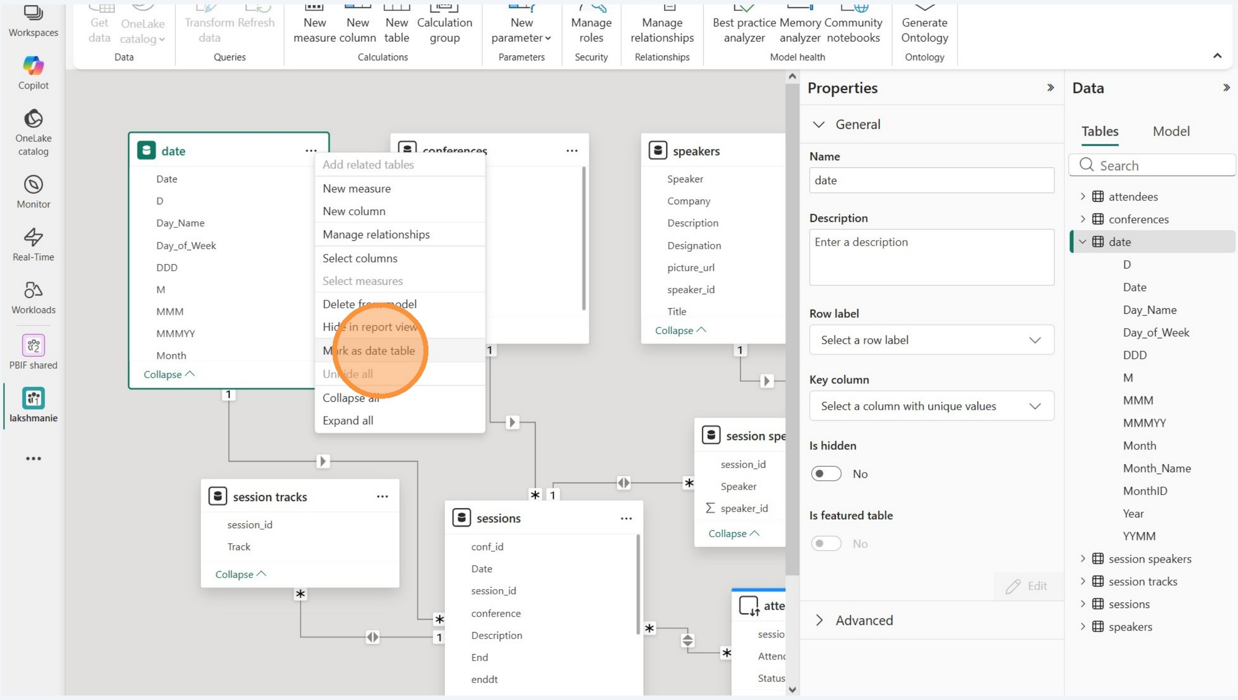
Click "Yes"



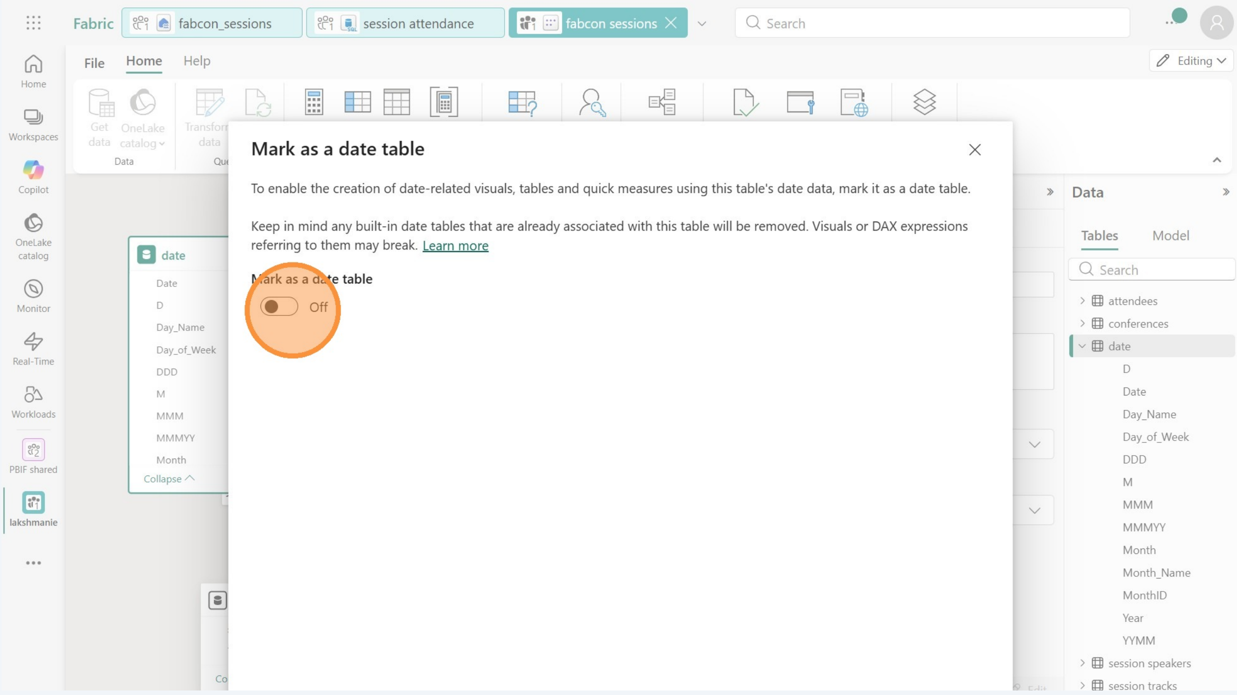
33 Now that the date table has a column that is of the date datatype, we can mark it is a date table. Click the 3 dots in the header of the date table.



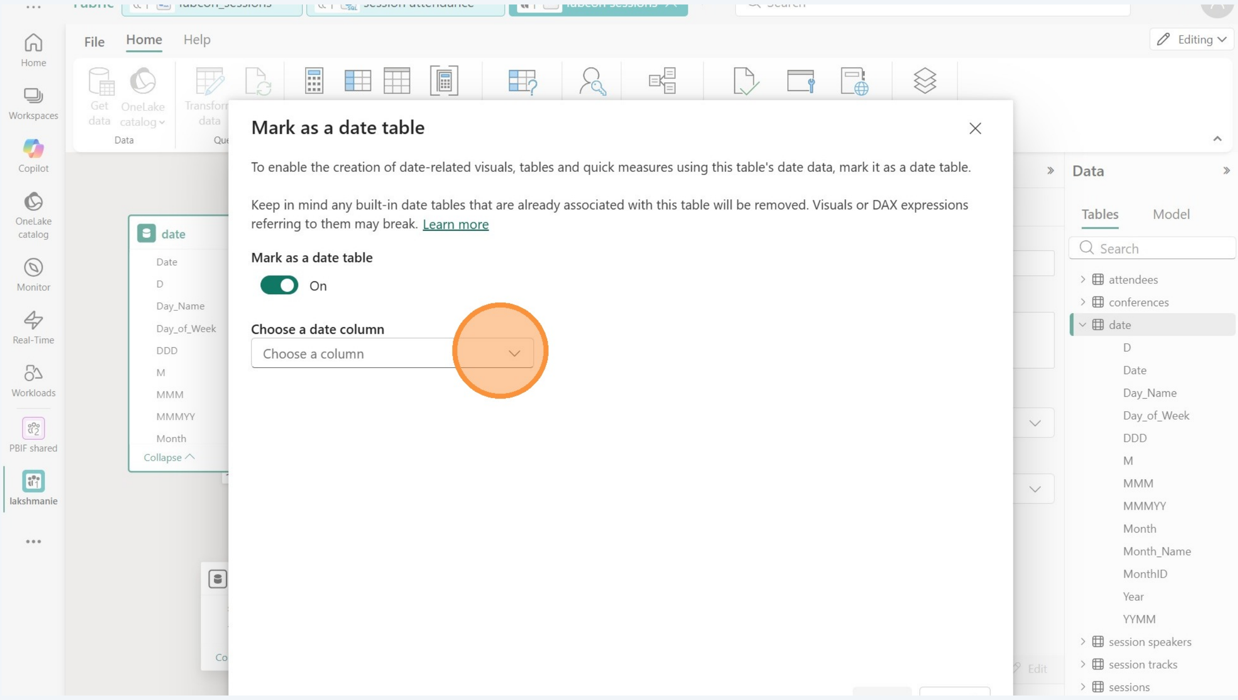
34 Click "Mark as date table"



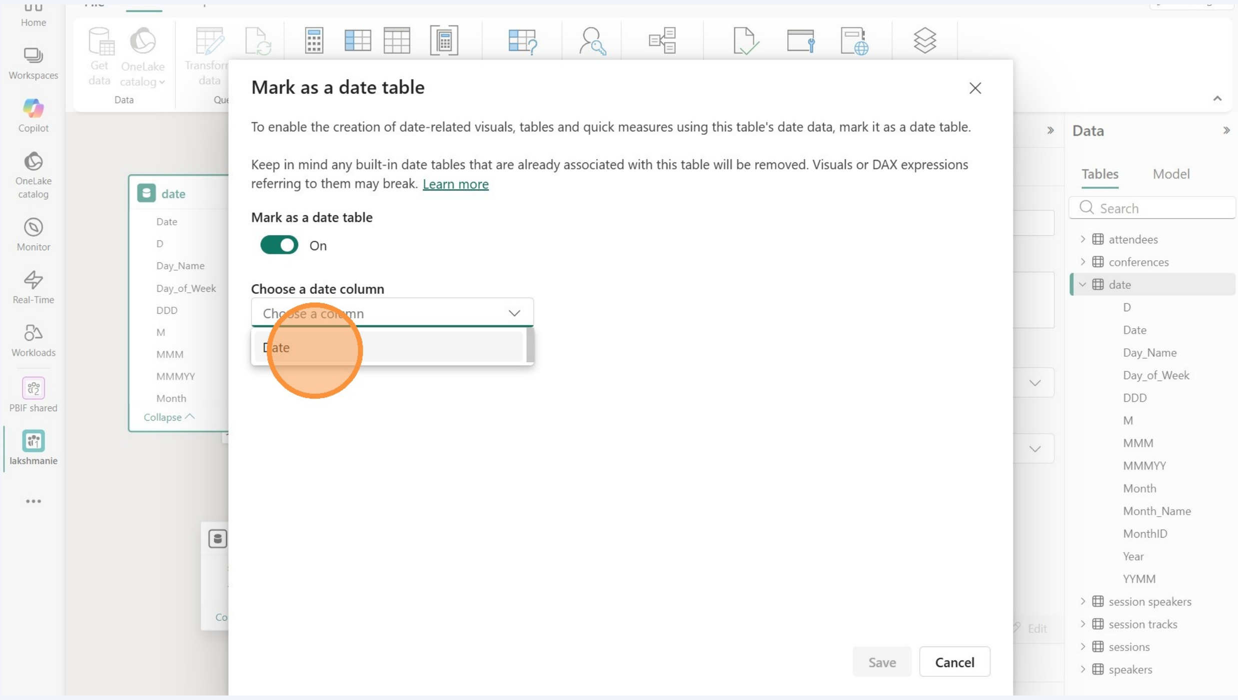
35 Turn the toggle switch to On.



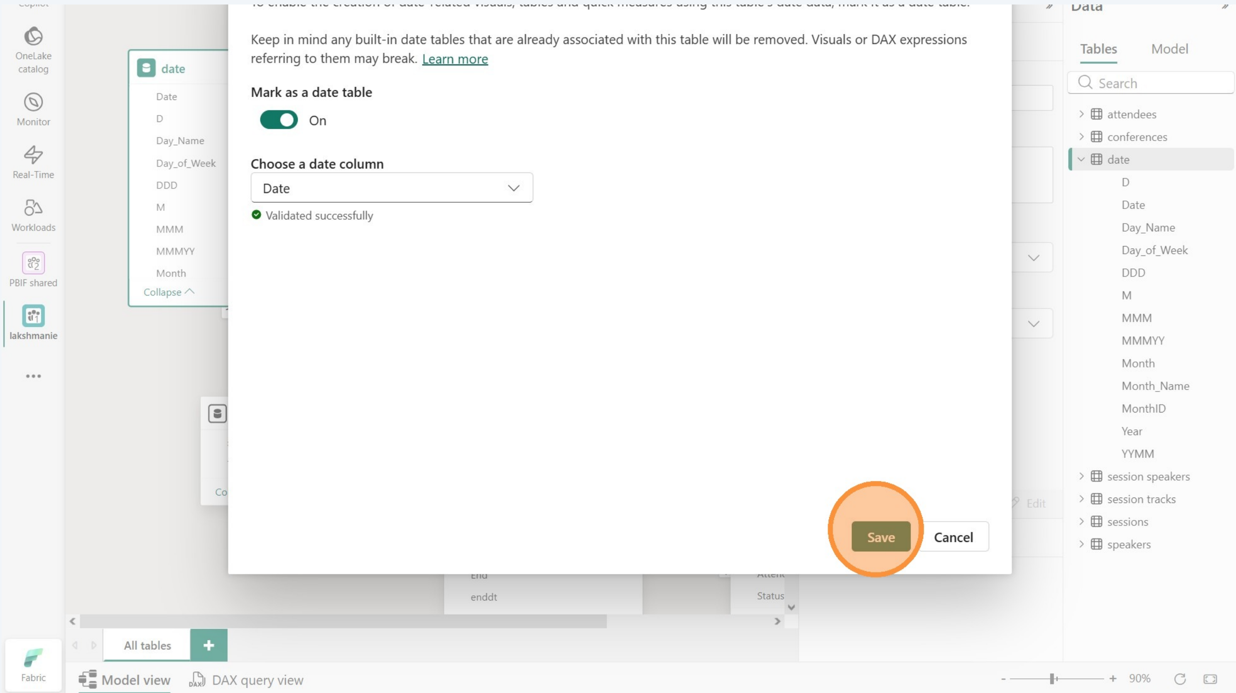
36 Click "Choose a column"



37 Click "Date"



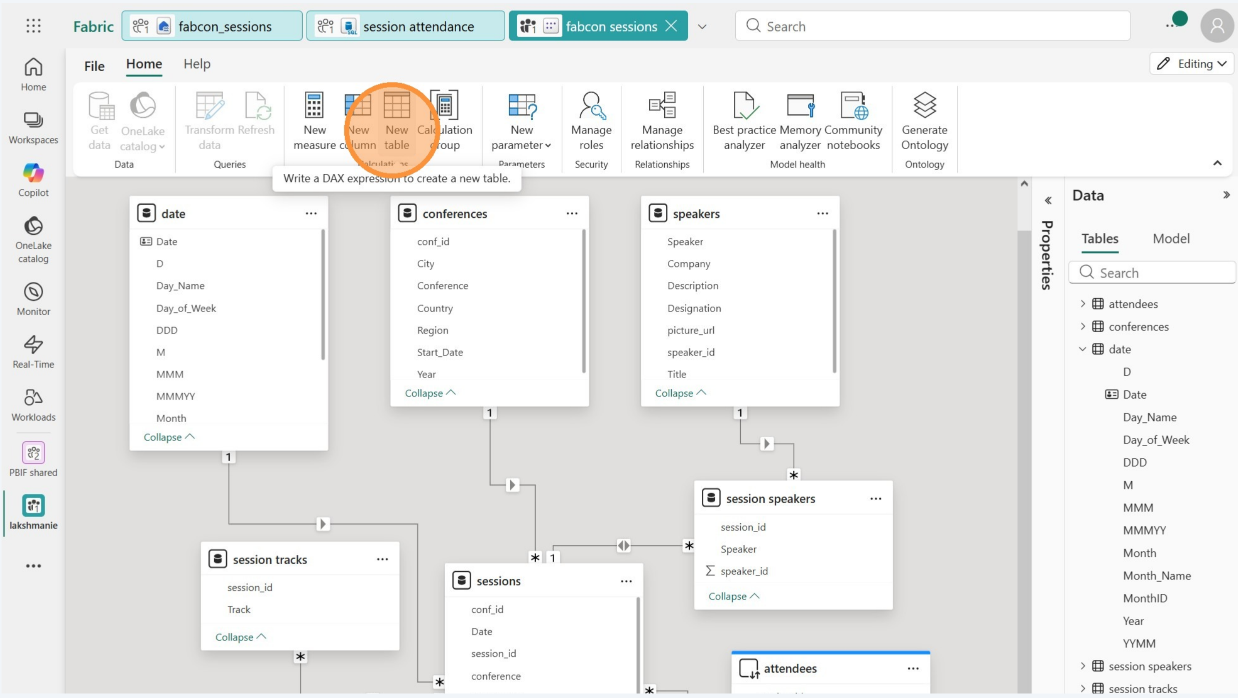
38 Click "Save"



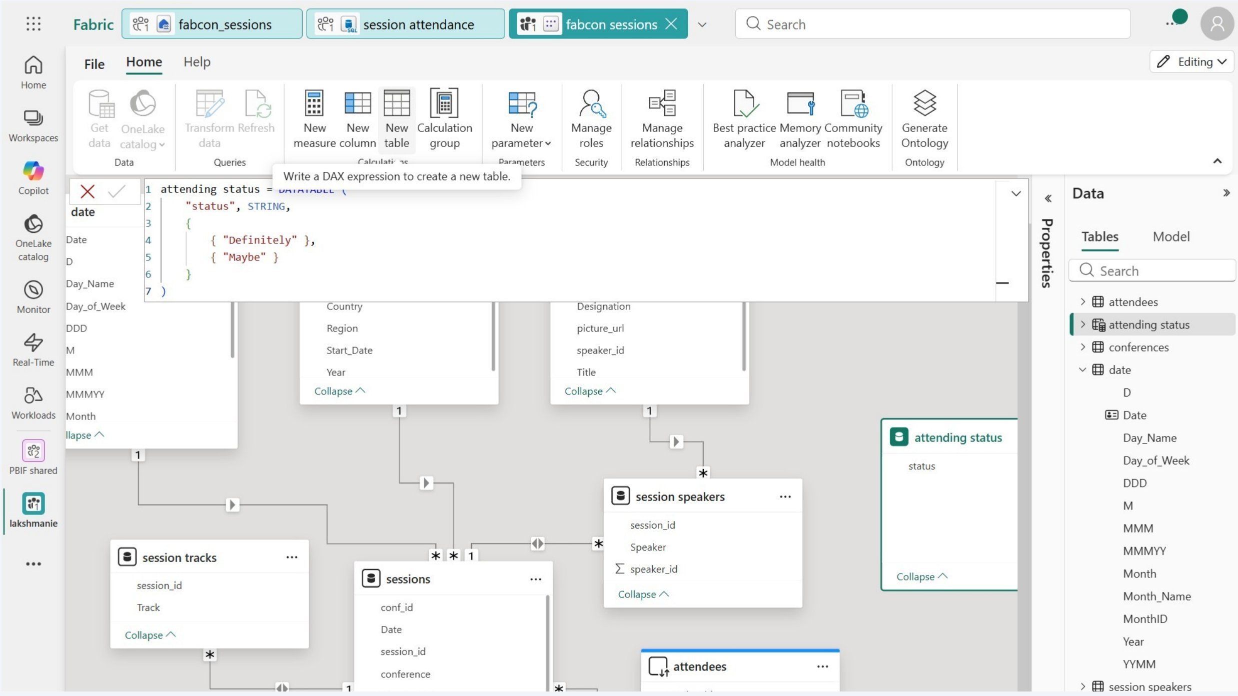
Create a table with DAX

39 We need one more table to populate a slicer for saving our session preferences. We want to be able to decide if we are "Definitely" or "Maybe" attending the sessions. So we'll create a table with just those two rows. No relationship will be required for this table.

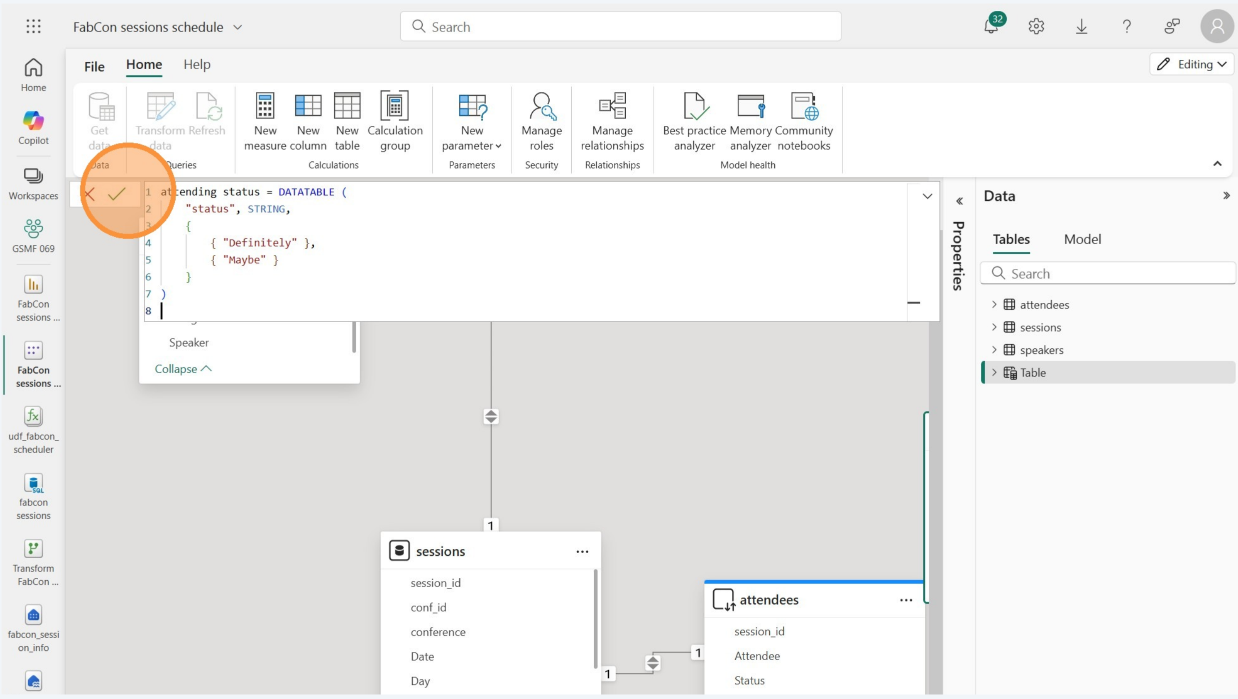
Click the "New table" button in the Home tab.



40 In the formula bar, paste the following:



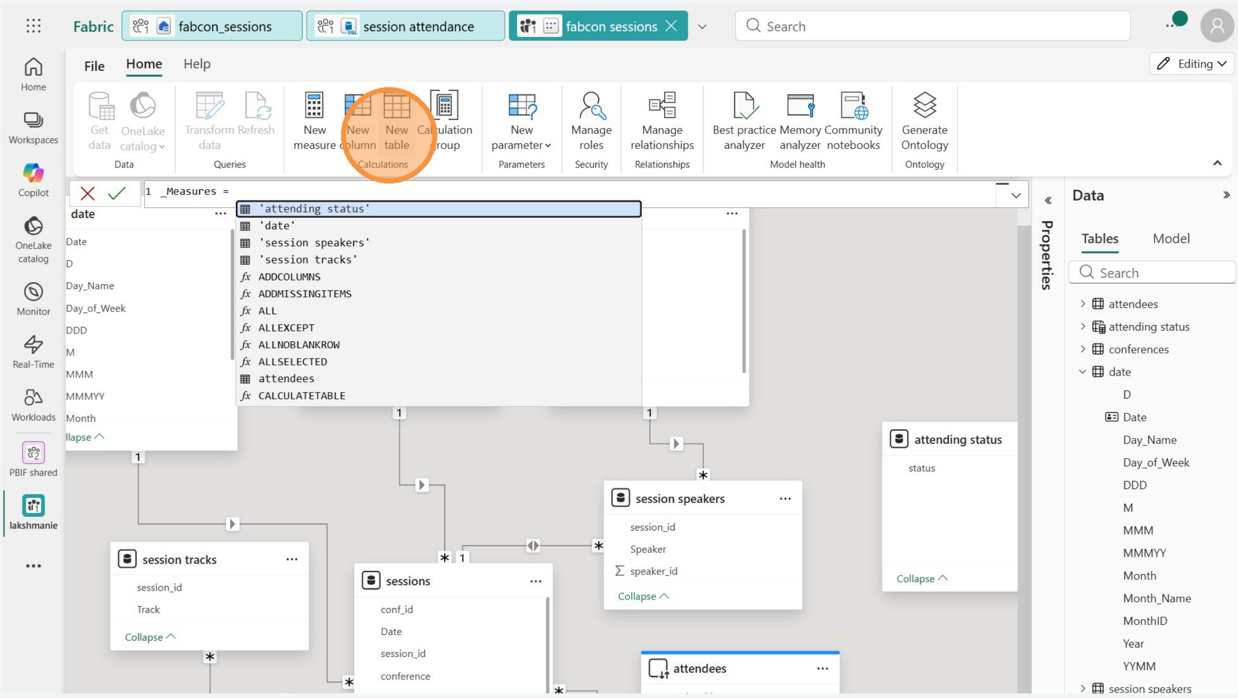
41 Click the green checkmark to the left of the formula bar to enter it.



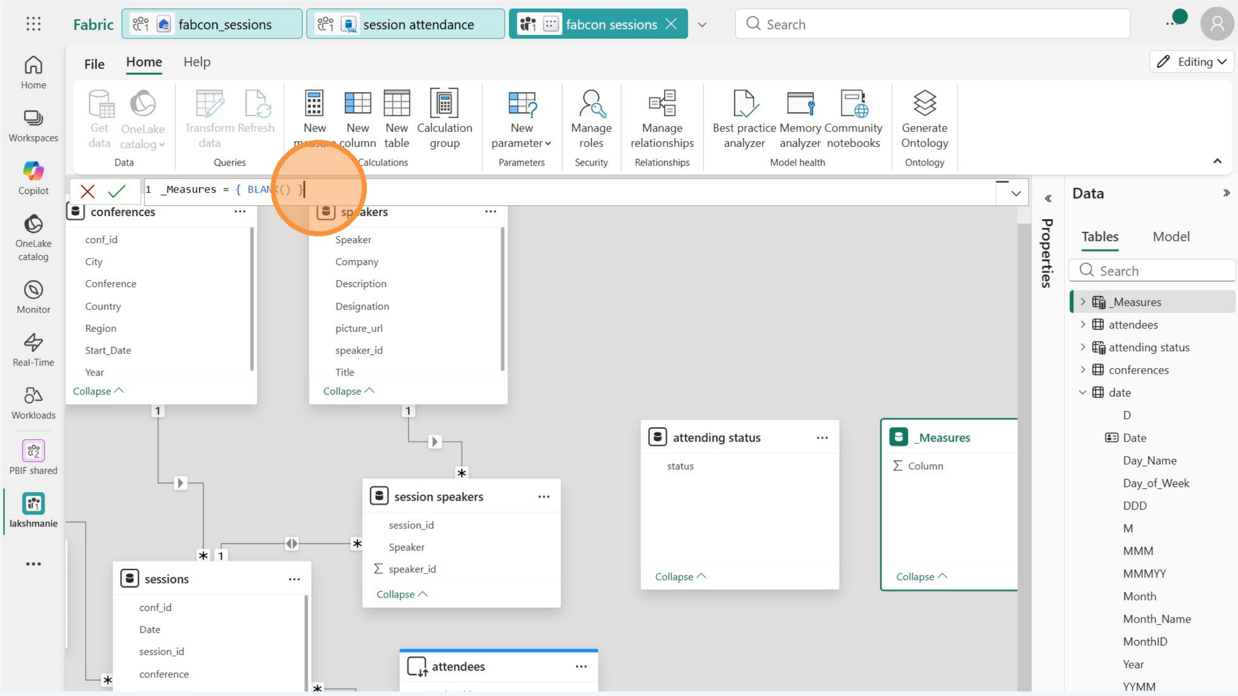
Create a measures table

Tip! You don't have to create a special table for measures, and if you go down a rabbit hole of measure table research, you'll find many strong opinions about whether to put measures in their own table. Today we will, but that doesn't mean you have to. You can make your own choices! You do you.

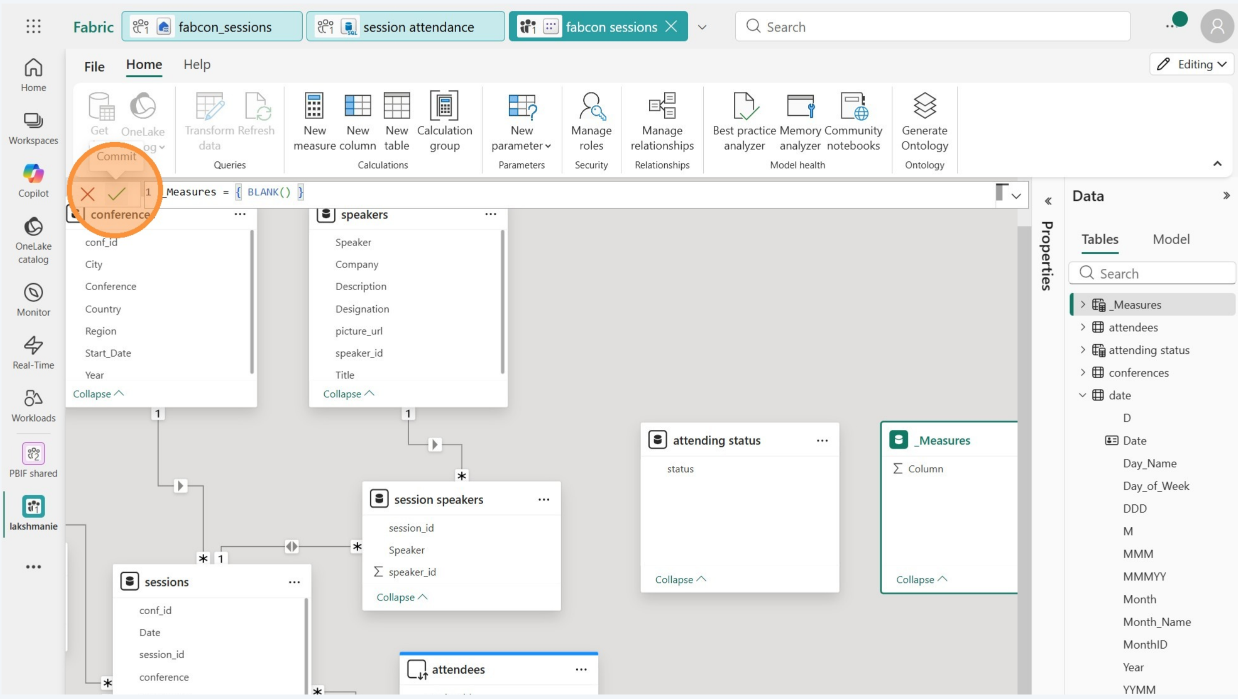
42 Click "New table"



43 Enter "_Measures = { BLANK() }"

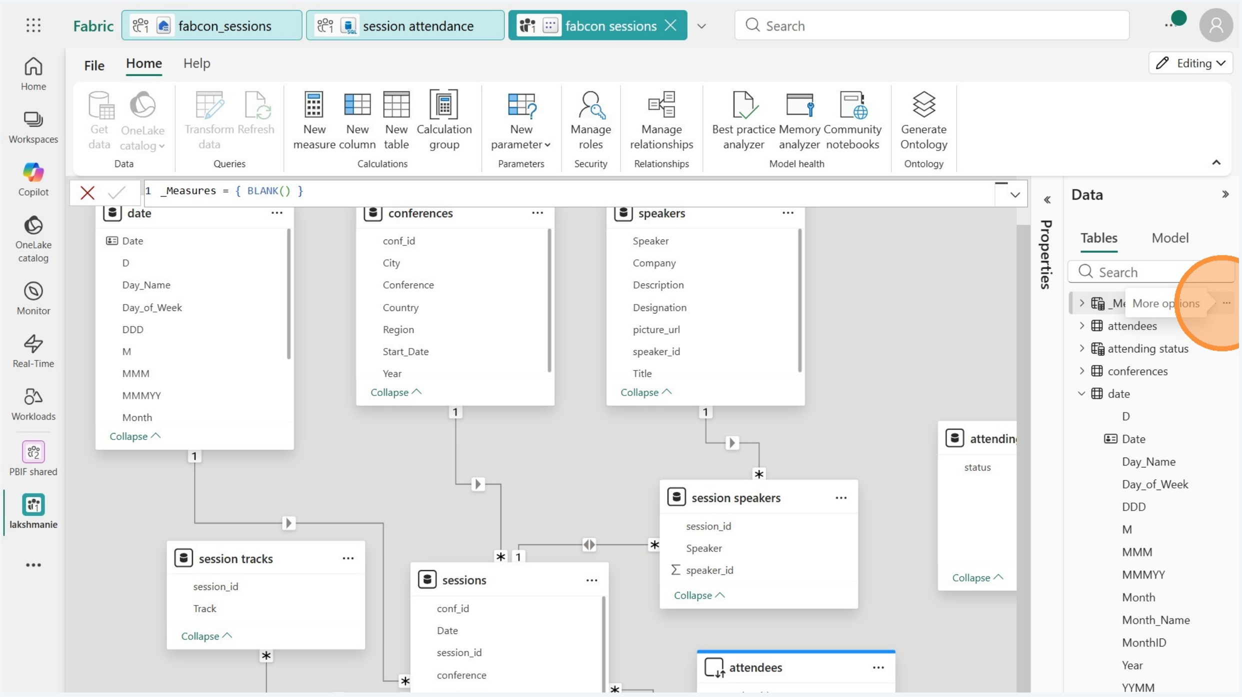


44 Click the green check mark to commit it.

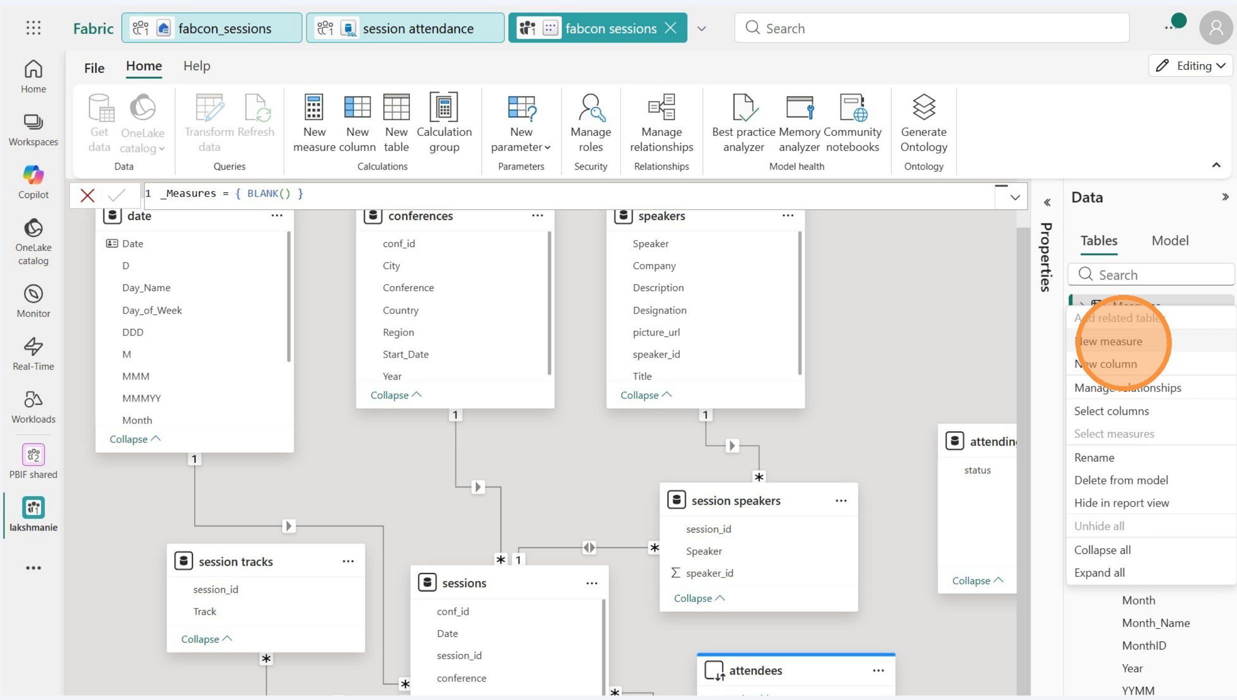


Create the measures

45 Click this the 3 dots next to the _Measures table.

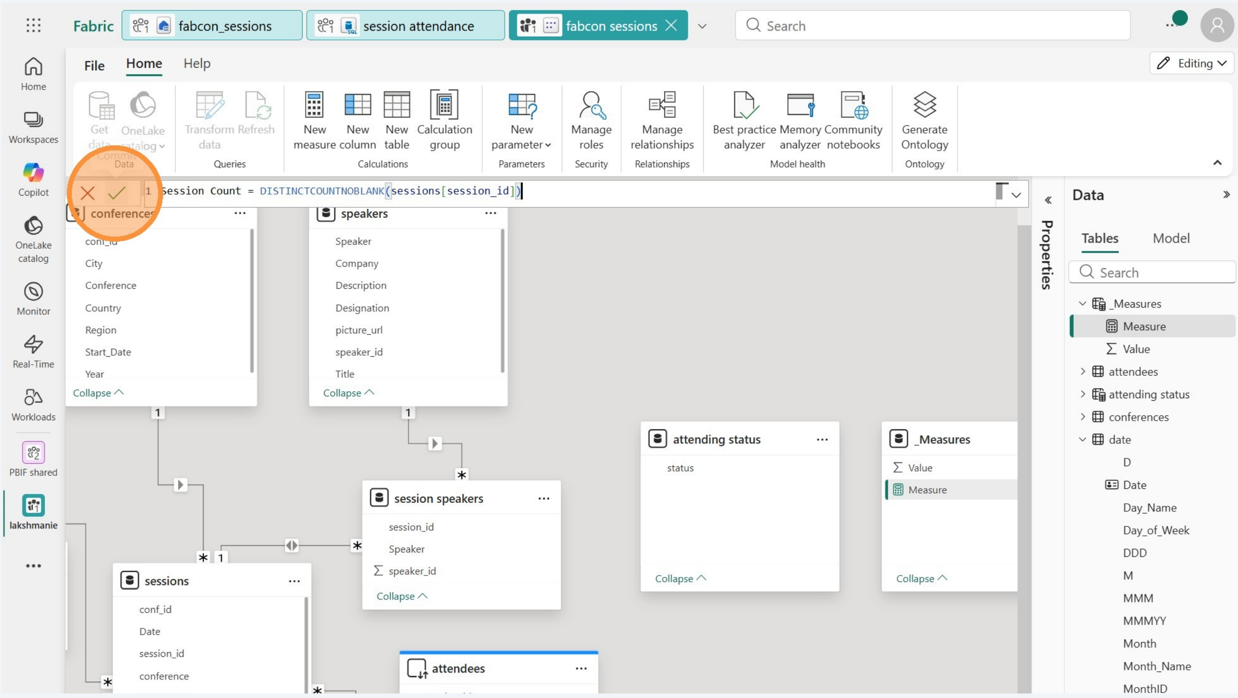


46 Click "New measure"



47 Let's create a measure for session count. Replace the default text with:

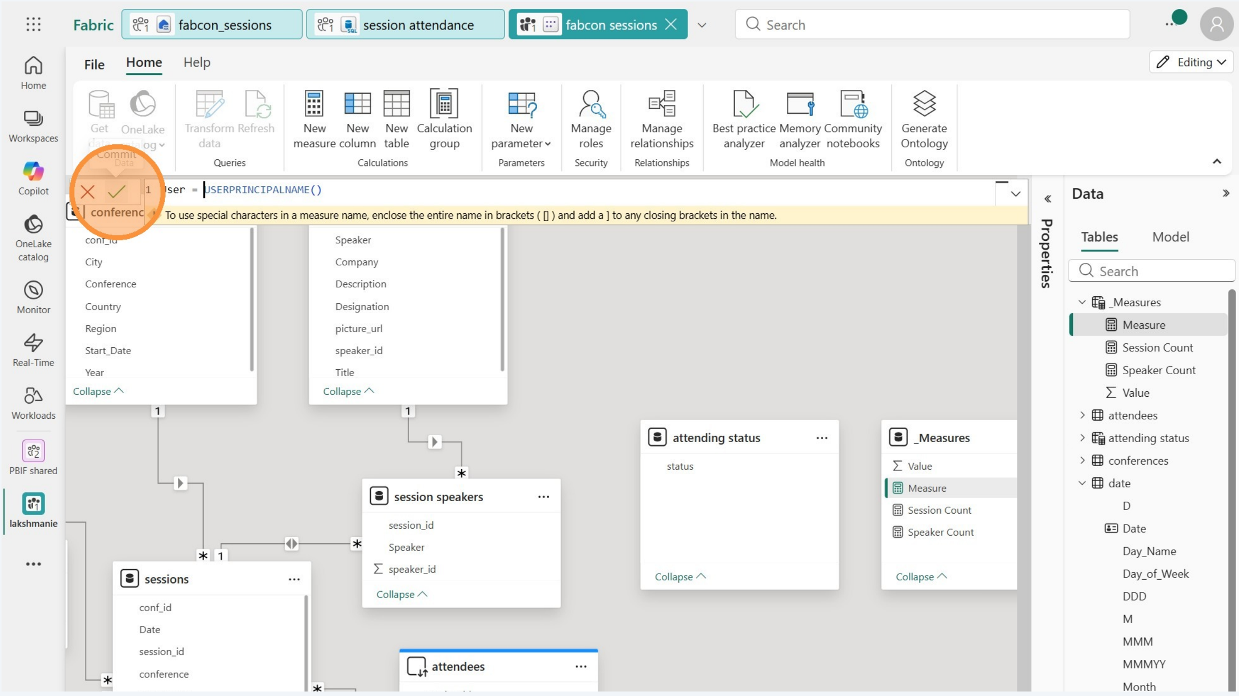
Click the green check box or just click "Enter."



48 Great job! Let's make another measure! Repeat the last steps to create another measure in the _Measures table.

Replace the default text with the following measure:
User = USERPRINCIPALNAME()

This measure will return the UPN of the user running the report. We'll use it for the attendees updates.



ⓘ Tip! In the measures you create below - ensure you're referring to the same table and column names that you created in your lakehouse and your SQL database. If you get a red squiggly on the (table[(columnname]) - update those with the correct table and column names that you have in your data!

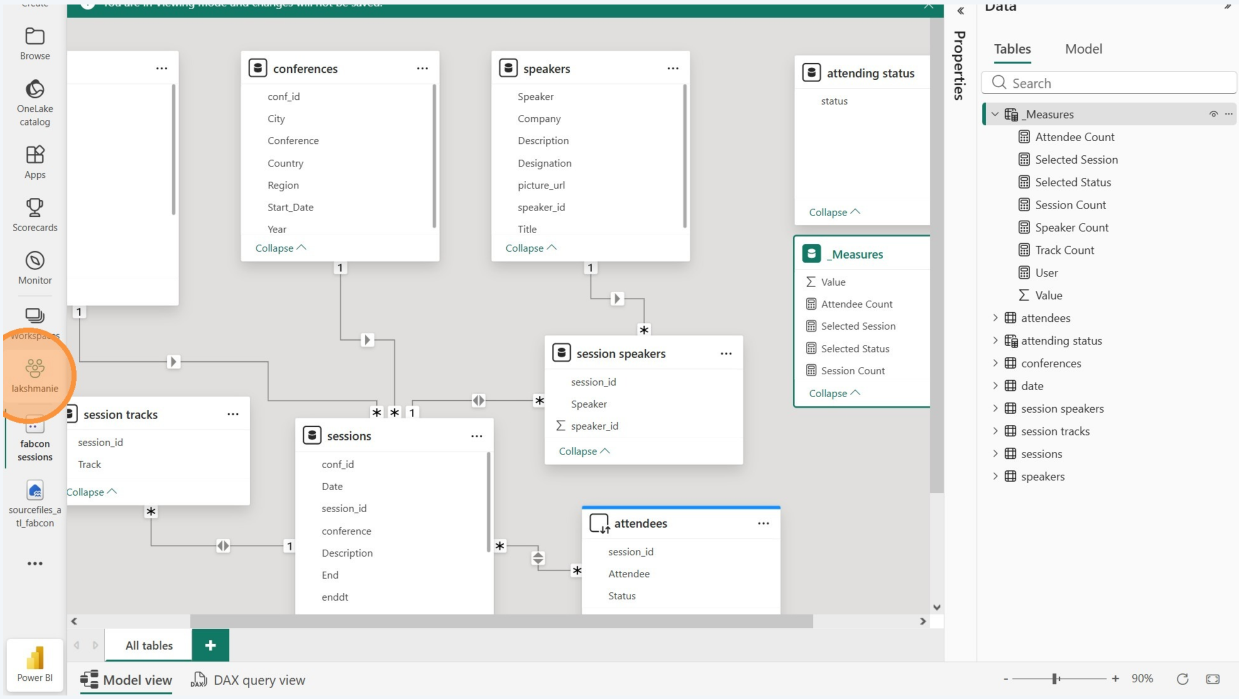
49 Do the same as above to create the following four measures:

[[Selected Session = SELECTEDVALUE(sessions[session_id])]]
[[Selected Status = SELECTEDVALUE('attending status'[status])]]
[[Speaker count = DISTINCTCOUNTNOBLANK(speakers[Speaker])]]
[[Attendee Count = **DISTINCTCOUNTNOBLANK**(attendees[Attendee])]]

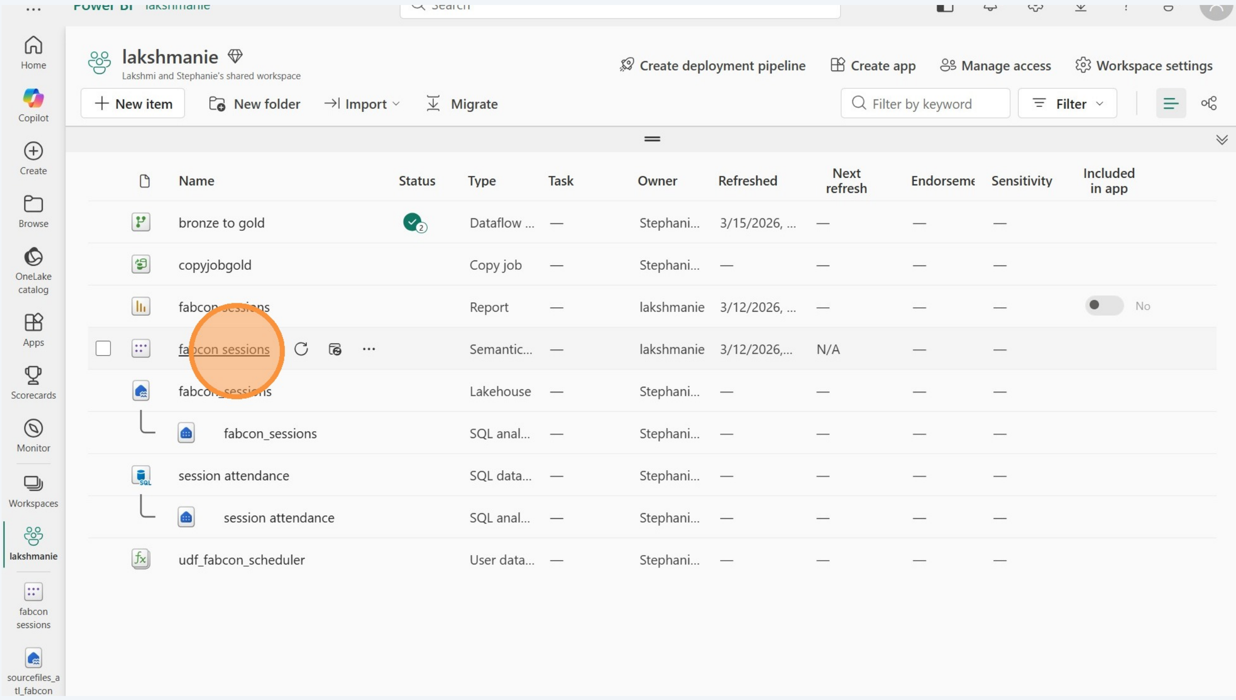
[[Track Count = DISTINCTCOUNT('session tracks'[Track])]]

We'll use the selected session and selected status measures for the User Defined Function we created earlier.

50 Make sure your model has the below relationships and measures. Then go back to your workspace.



51 Click your semantic model to view info about it.



52

If you expand the _Measures table you can see the measures you created. You can scroll through all the tables.